Software Engineering Department

Braude College of Engineering

Capstone Project Phase B (61999)

CASEL in Higher Education:

AI-Guided Socratic Dialogue for Developing Social-Emotional Learning (SEL) Skills in Higher Education

CODE: 26-1-R-5

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Link to GitHub:

https://github.com/NahlaAboromi/final_project

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Abstract

In recent years, there has been growing recognition of the importance of Social Emotional Learning (SEL) among higher education students, due to its influence on emotional well being, sense of belonging, and persistence in academic studies. Numerous studies indicate that SEL contributes to academic success and emotional well being, while strengthening emotional regulation and institutional belonging. However, many students still lack access to practical, accessible, and non judgmental opportunities to practice SEL independently of teaching staff or formal academic courses. This gap highlights the need for innovative models that do not require additional instructional resources and do not expose students to emotional vulnerability, thereby enabling more equitable responses to students' social emotional needs. This study evaluated the effectiveness of a brief AI guided Socratic dialogue in promoting Social Emotional Learning (SEL) competencies among higher education students without instructor intervention or evaluative or judgmental feedback. A randomized controlled experiment was conducted using the CASELy platform, with N=174 higher education students recruited from nine academic institutions, 83.9% of whom were recruited from Braude College of Engineering. Participants were randomly assigned to one of four groups: three experimental groups (A, B, C) and one control group (D). All groups completed a social emotional simulation scenario and received an AI generated CASEL analysis. The experimental groups additionally engaged in a short AI guided Socratic dialogue, whereas the control group proceeded directly without a conversational component. All participants completed validated pre and post intervention SEL measures and the User Experience Questionnaire Short Version (UEQ-S) at the end of the intervention. The sample comprised two cohorts: Advanced Students (n=101), recruited voluntarily across multiple institutions, and First Year Students (n=73), recruited through the Engineering Fundamental Skills course at Braude College of Engineering. Quantitative findings revealed significant pre to post improvements in SEL competencies among experimental participants across both cohorts, whereas no significant changes were observed in the control group. First year students demonstrated significant gains across all five CASEL domains (all $p < .001$), while advanced students showed smaller but statistically significant improvements in overall SEL and Social Awareness ($p < .05$). Experimental participants also rated the system significantly higher than control participants across all UEQ-S dimensions, indicating that the Socratic dialogue transformed the experience from a passive assessment into an engaging reflective interaction. Qualitative analysis of dialogue transcripts revealed consistent thematic patterns across both cohorts, including shifts from automatic emotional reactions toward deliberate value aligned decision making, reframing of help seeking as strength, and recognition of rationalization under pressure, with advanced students demonstrating deeper interpretative reflection compared to first year students. The findings demonstrate that a brief, anonymous, and instructor independent AI guided Socratic dialogue effectively promotes SEL competencies in higher education, with stronger effects observed among students at the beginning of their academic journey. These findings suggest that AI guided Socratic dialogue offers a scalable, equitable, and pedagogically viable approach for embedding SEL as a continuous experiential process within everyday academic experiences without requiring additional instructional resources or professional training.

Keywords: Social Emotional Learning (SEL),Socratic Dialogue,Artificial Intelligence (AI),Higher Education,CASEL

1. Introduction

Higher education students face growing emotional and psychological challenges, including anxiety, depression, loneliness, and diminished sense of belonging, that directly affect academic performance, motivation, and well-being (Conley & Donahue-Keegan, 2024; Simion, 2023). Social Emotional Learning (SEL), grounded in the CASEL framework of five core competencies (self awareness, self management, social awareness, relationship skills, and responsible decision making), has demonstrated consistent positive effects on students' mental health, emotional regulation, and academic success (Takizawa et al., 2024; Johnson, 2020; Hareli et al., 2025). Despite this evidence, most existing SEL interventions rely on instructor availability or formal course structures, leaving many students without accessible, non judgmental opportunities for SEL practice. Research on AI based Socratic approaches, which offer personalized, scalable, and anonymous support, remains limited (Wang D. & Ishak, 2025). This study therefore asks: **How does an AI based Socratic dialogue influence the development of SEL skills among higher education students?**

2. Literature Review

2.1 SEL in the academy

Research consistently highlights the central role of Social and Emotional Learning (SEL) in higher education. Conley argues that systematic SEL integration strengthens students' mental well being, sense of belonging, and academic achievement while fostering equity (Conley & Donahue-Keegan ,2024). Aligned with the CASEL model's five core competencies — self-awareness, self-management, social awareness, relationship skills, and responsible decision making, SEL functions as a preventive framework reinforcing resilience and academic motivation. Empirical studies support its long term impact: (Hareli et al. ,2025) found sustained benefits including emotional resilience and interpersonal skills months after an SEL-integrated course, suggesting that SEL is a long-term developmental process influencing professional transitions. (DeSensi ,2024) further emphasizes that SEL should be embedded within course design through collaborative learning and project based activities that cultivate belonging, emotional regulation, and resilience. Wang's systematic review (Wang & Ishak, 2025) identifies the absence of frameworks for capturing how SEL competencies evolve in AI-mediated environments, a gap directly addressed by the current study.

2.2 SEL Assessment and Intervention Outcomes

Validated questionnaires play a central role in measuring SEL competencies in higher education. (Ahmad and Singh ,2025) found that among 300 undergraduates, students showed strengths in social awareness and relationship skills but gaps in self-management and responsible decision-making. (Ren et al. ,2025) demonstrated that SEL significantly improved mental health among 600 university students, with self-awareness and self-management particularly linked to reduced psychological distress. (Johnson ,2020) further confirmed, through a pre-test/post-test controlled design, that structured SEL programs improve emotional regulation, interpersonal skills, and academic stress management. These findings provide support for the use of a pre-test/post-test design when evaluating SEL interventions in higher education.

2.3 AI Based Approaches to SEL

Despite the demonstrated benefits of traditional SEL interventions, research on AI guided, Socratic style approaches remains limited (Wang & Ishak, 2025). Such approaches offer unique advantages: scalability, accessibility, anonymity, and personalization without requiring instructor availability. The current study directly addresses this gap by examining whether a brief AI based Socratic dialogue can promote SEL competencies among higher education students in a controlled, instructor independent setting.

3. Pilot Study and Preliminary Evaluation

3.1 Initial Field Experiment

The initial experimental run was conducted within the course Engineering Fundamental Skills (Course 251961), with participation from two instructors' groups: 5 students from Dr. Dana Fischer Shahor's group and 11 students from Dr. Magi Ben Yehuda's group (N=16 total).

3.2 Identified Challenges and System Improvements

Following the session, a systematic examination of logs and admin dashboard activity revealed two main categories of challenges, along with the corrective measures applied to each. **Table 1** presents the insights:

Table 1: Identified Challenges and Implemented Solutions

Category	Identified Challenge	Implemented Solution
AI Model Limitations	Token limits caused the process to terminate prematurely for some participants	Expanded token capacity and introduced a fallback mechanism to a secondary AI model in cases of overload
UX and UI: Dialogue Skipping	Some participants exited immediately after the CASEL analysis, interpreting the process as complete	Enhanced navigation clarity with explicit emphasis on the importance of the dialogue stage
UX and UI: Visual Guidance	Some participants skipped the Socratic dialogue due to insufficient visual cues	Added a progress indicator throughout the Socratic dialogue and improved visual positioning of the Socratic bot

Note. All identified challenges were resolved prior to the main experiment. A controlled post-fix validation session (N=10) confirmed smooth system performance across all components, with no critical issues detected.

4. Experimental design

4.1 Research Design

This study employed a randomized controlled experiment to examine whether a brief AI guided Socratic dialogue supports the development of Social-Emotional Learning (SEL) competencies among higher education students. Participants were automatically and randomly assigned by the CASELy platform to one of four groups in a 1:1:1:1 ratio: three experimental groups (A, B, C) and one control group (D). All groups completed a validated pre-intervention SEL questionnaire (Ahmad & Singh, 2025), engaged with a social-emotional simulation scenario, and received an AI-generated CASEL competency analysis. The experimental groups then participated in a guided Socratic dialogue with the AI system, received a personalized AI summary of their reflective process, completed the post-intervention SEL questionnaire (Ahmad & Singh, 2025), responded to two open-ended reflection questions, and finally completed the User Experience Questionnaire UEQ-S (Schrepp et al., 2017). The control group proceeded directly from the

CASEL analysis to the post-intervention SEL questionnaire and UEQ-S, without any conversational component.

4.2 Participants

A total of N=174 higher education students participated in the study, recruited from nine different academic institutions, including Braude College of Engineering, Technion Israel Institute of Technology, Hebrew University of Jerusalem, Ben Gurion University of the Negev, Bar Ilan University, Arab American University, Jerusalem Multidisciplinary College, Yezreel Valley College, and Gordon Academic College. However, Braude College of Engineering served as the leading institution, contributing 146 of the 174 participants (83.9%), while the remaining 28 participants (16.1%) were distributed across the other eight institutions. The sample comprised two cohorts: Advanced Students (n=101), recruited across all nine institutions, and First Year Students (n=73), recruited through the Engineering Fundamental Skills course (Course 251961) at Braude College. All were randomly assigned to one of four research groups in a 1:1:1:1 ratio. Full demographic details and group assignment are presented in **Table 2**.

Table 2: Participant Demographics and Group Assignment

category	details	Advanced Students (n=101)	Year 1 Students (n=73)	Total (N=174)
Total Participants	All	101	73	174
Gender Distribution				
Gender	Male	48	40	88
	Female	52	33	85
	Other / Not Specified	1	0	1
Institution				
Institution	Braude College	73	73	146
	Other Institutions	28	0	28
Research Group Assignment				
Research Group	Group A (Experimental)	22	21	43
	Group B (Experimental)	25	18	43
	Group C (Experimental)	27	16	43
	Group D (Control)	27	18	45

Note. Year 1 Students participated through the Engineering Fundamental Skills course (Course 251961). All participants were anonymously assigned system generated UUIDs.

4.2.1 Age Distribution

Participants represented a broad range of age groups (18–36+) and academic fields, with the majority falling between 18 and 26 years of age (85.1%), as detailed in **Table 3**.

Table 3: Age and Institutional Distribution (N=174)

Age Range	Braude	Outside Braude	Total
18–22	62 (35.6%)	10 (5.7%)	72 (41.4%)
23–26	62 (35.6%)	14 (8.0%)	76 (43.7%)
27–30	17 (9.8%)	4 (2.3%)	21 (12.1%)
31–35	4 (2.3%)	0 (0.0%)	4 (2.3%)
36+	1 (0.6%)	0 (0.0%)	1 (0.6%)

4.3 Measures

4.3.1 Social Emotional Learning (SEL) Questionnaire

SEL competencies were assessed using the validated SEL Skills Survey (Ahmad & Singh, 2025), administered as both a pre- and post-intervention measure. The instrument comprises 20 items distributed across the five CASEL domains (four items per domain), rated on a four-point Likert scale from 1 (*Strongly Disagree*) to 4 (*Strongly Agree*). Domain and overall SEL scores are computed as item means. Score interpretation follows the thresholds in **Table 4**. The full instrument is provided in Appendix A and is publicly available at [LINK].

Table 4: Interpretation Data (Adapted from Ahmad & Singh, 2025)

Weight	Scale Range	Description	Interpretation
4	3.25 – 4.00	Strongly Agree	High Level
3	2.50 – 3.24	Agree	Average Level
2	1.75 – 2.49	Disagree	Low Level
1	1.00 – 1.74	Strongly Disagree	Very Low Level

4.3.2 User Experience Questionnaire Short Version (UEQ-S)

User experience was assessed using the UEQ-S (Schrepp et al., 2017), an eight-item semantic differential scale. Raw scores (1-7) are transformed to a standardized −3 to +3 scale, yielding two dimensions: Pragmatic Quality (items 1-4) and Hedonic Quality (items 5-8), with an overall score computed as the mean of all eight items **(Table 5)**. The full instrument is provided in Appendix B and is publicly available at [LINK].

Table 5: UEQ-S Scoring Structure and Dimensions (Adapted from Schrepp et al. ,2017)

Dimension	Items	Score Range	Description
Pragmatic Quality	Items 1–4	−3.00 to +3.00	Reflects perceived usability, clarity, efficiency, and task support
Hedonic Quality	Items 5–8	−3.00 to +3.00	Reflects perceived novelty, stimulation, originality, and engagement
Overall User Experience	Items 1–8	−3.00 to +3.00	Represents overall subjective evaluation integrating pragmatic and hedonic aspects

4.4 Simulation Scenarios

Four simulation scenarios were developed for this study, one assigned to each research group (A, B, C, and D). All scenarios were structurally equivalent in length, complexity, and the five CASEL competencies they activated, ensuring that any observed differences between groups could be attributed to the intervention rather than to scenario format or difficulty. Each scenario described an authentic academic social-emotional situation and was adapted from publicly available sources to ensure activation of all five CASEL competencies: self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. All scenarios were authored and validated by an expert in student development and higher education pedagogy prior to use in the study. The full set of scenarios is publicly available at [LINK].

4.5 Procedure

The experiment was administered entirely online through the CASELy platform. **Figure 1** illustrates the complete procedural flow followed by participants throughout the study. All participants completed an anonymous demographic form and were randomly assigned to one of four conditions before beginning. Experimental groups (A, B, C) completed the pre-intervention

SEL questionnaire, engaged with their assigned simulation scenario, received an AI generated CASEL analysis, participated in the Socratic dialogue, received a personalized AI summary, completed the post-intervention SEL questionnaire, responded to two open-ended reflection questions, and completed the UEQ-S. The control group (D) followed the same procedure with the exception that no Socratic dialogue, AI summary, or reflection questions were presented. The full analysis code is publicly available at [LINK].

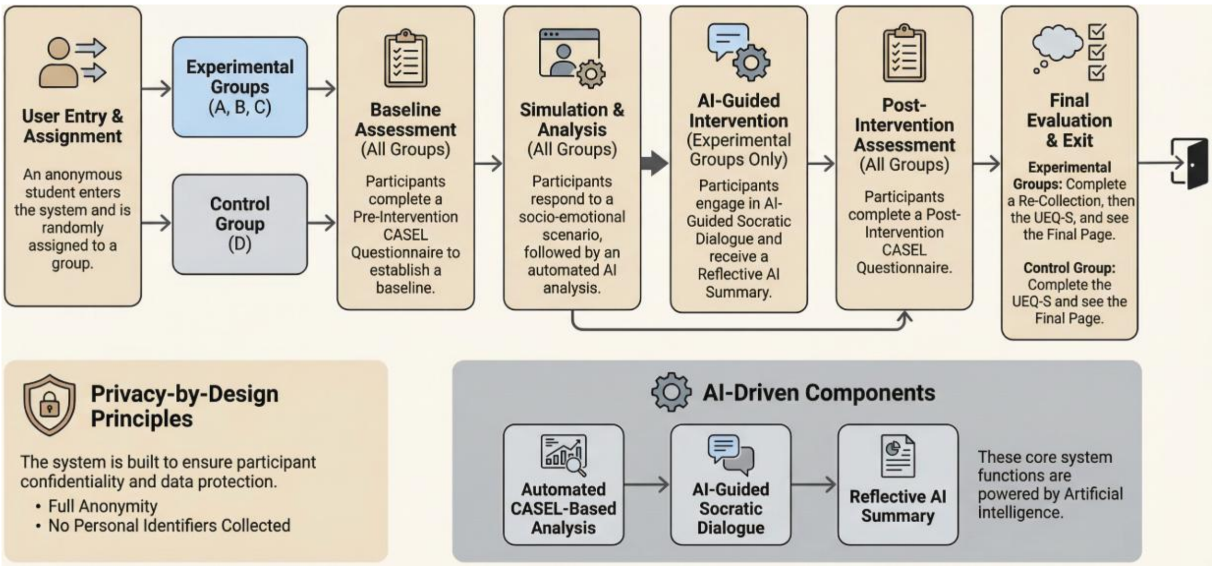

Figure 1. Overview of the experimental procedure and system workflow for experimental and control groups

4.5.1 System Infrastructure

The experiment was administered through the CASELy platform, deployed on a dedicated cloud server hosted on Contabo, running Microsoft Windows Server 2025 on an x64-based environment with 8GB RAM. The backend was implemented using Node.js (v25.8.0) and managed with PM2 to ensure continuous operation and automatic recovery. The system exposed standard web services via HTTP (port 80) and HTTPS (port 443), with the backend API running on port 5000. A reverse proxy based on Caddy was used to provide secure communication and automatic certificate management. This deployment ensured a stable, scalable, and secure environment, enabling reliable access and consistent system performance throughout the experimental study.

4.6 Ethical Considerations

This study was reviewed and approved by the Research Ethics Committee of Braude Academic College of Engineering (Approval No. 2026-05, granted 15 March 2026). All participants took part voluntarily and were informed on the entry page that participation was anonymous, that no identifying information was collected, and that responses were used solely for research purposes. No names, email addresses, student ID numbers, or any other personal identifiers were collected at any stage; all data were stored under system generated anonymous UUIDs. Demographic data were limited to four non-identifying fields (field of study, current semester, age range, and gender) sufficient for group comparisons while preserving full anonymity. Participants could discontinue at any time without consequence. Because the study involved no deception, no sensitive personal data, and no risk of harm, a full written informed consent form was not required; the platform's entry-page privacy statement served as the disclosure mechanism, consistent with the ethics committee's conditions of approval.

5. Results

5.1. First Year Student task completion

Table 6 presents completion and engagement metrics for first year students across the experimental and control groups, summarizing participation across the different stages of the study and interaction with the Socratic chatbot.

Table 6: Completion and Engagement Metrics of First Year Students

Stage / Metric	Group A	Group B	Group C	Group D (ctrl)	Total exp.
Group size					
N in group	21	18	16	18	55
Core completion (experimental)					
Completed PRE	21/21	18/18	15/16	18/18	54/55
Completed simulation	20/21	18/18	15/16	17/18	53/55
Trivial simulation answer	~2	~1	~1	~2	~4
AI chat time > 0 min	~19	~16	~13	-	48/55
Received AI summary	~20	~17	~13	-	50/55
Completed POST	~18	~16	~13	16/18	47/55
Completed reflection (2 Q)	~17	~15	~13	-	45/55
Completed UEQ	~17	~15	~12	16/18	44/55

Notes. *Trivial answers = single letter or single word simulation responses ("b", "ג", "n"). 7 participants had 0 minute AI chat time.*

5.2 Advanced Year Student task completion

All 101 advanced students completed the study with no participant dropouts. **Table 7** summarizes the completion and engagement metrics across the experimental and control groups, showing consistently high completion rates throughout the study.

Table 7: Completion and Engagement Metrics of Advanced Students

Stage / Metric	Group A	Group B	Group C	Group D (ctrl)	Total exp.
Group size					
N in group	22	25	27	27	74
Core completion (experimental)					
Completed PRE	22/22	25/25	27/27	27/27	74/74
Completed simulation	22/22	25/25	27/27	27/27	74/74
Trivial simulation answer	0	0	0	0	0
AI chat time > 0 min	22	25	27	-	74/74
Received AI summary	22	25	27	-	74/74
Completed POST	21/22	24/25	27/27	25/27	72/74
Completed reflection (2 Q)	21	25	27	-	73/74
Completed UEQ	21/22	25/25	27/27	26/27	73/74

5.3 Comparison of Completion Rates Between Cohorts

Advanced students demonstrated near perfect completion across all stages (100% dropout free, 97–100% per stage), whereas first year students showed greater variability, with stage level completion ranging from approximately 78% to 98%. Several factors may explain this difference. Although participation was voluntary in both cohorts, advanced students were recruited through voluntary calls across multiple institutions, whereas first year students were recruited within the context of a mandatory course. In addition, first year students were at the beginning of their academic studies and may have been less familiar with reflective learning activities and AI mediated dialogue. These differences in recruitment context and academic experience may have influenced engagement with the intervention. Additionally, trivial simulation responses and zero AI chat time were observed exclusively among first year students, suggesting that a small subset engaged only superficially with the task. Despite these differences, overall completion rates among first year students remained high.

5.4 Observed Task Completion Times

Table 8 presents the observed mean completion times per study component, calculated from system-recorded timestamps across all participants who completed the full experimental procedure. Experimental participants spent an average of 24.29 minutes completing the full study, compared to 14.48 minutes for control participants. The difference in total time is largely attributable to the Socratic Dialogue stage (7.37 minutes), which was present only in the experimental condition (see Table 8).

Table 8: *Observed Completion Times by Study Component (in minutes)*

Component	Experimental Groups (A, B, C)	Control Group (D)
Pre-SEL Questionnaire	2.34 min	1.95 min
Simulation Scenario & Written Response	5.55 min	6.99 min
Socratic Dialogue with AI Coach	7.37 min	-
Post-SEL Questionnaire	1.29 min	1.45 min
UEQ-S System Evaluation	0.66 min	0.69 min
Total	**24.29 min**	**14.48 min**

6. Quantitative analysis

6.1 Questionnaires

6.1.1 Advanced Students: User Experience (UEQ-S) and SEL Outcomes: Experimental vs. Control

The UEQ-S results revealed statistically significant differences between the experimental and control groups across all three dimensions. Experimental participants rated the system significantly higher than control participants on Pragmatic Quality ($t = 2.51$, $p = .014$, Cohen's $d = 0.57$), Hedonic Quality ($t = 3.23$, $p = 0.002$, Cohen's $d = 0.74$), and Overall Score ($t = 2.96$, $p = 0.004$, Cohen's $d = 0.68$), with all three differences reflecting medium effect sizes. Overall, the experimental group perceived the platform as more usable, more engaging, and more stimulating than the control group.

A likely explanation for this difference is the Socratic dialogue component. Experimental participants engaged in an interactive, personalized AI guided conversation, whereas control participants received only a static CASEL analysis with no conversational element. The dialogue appears to have transformed the experience from a passive task into an active, reflective interaction, making the platform feel more meaningful and engaging. This interpretation is further supported by the particularly large effect size observed for Hedonic Quality (Cohen's $d = 0.74$), which captures novelty and stimulation.

Beyond user experience, the impact of the intervention on social-emotional learning (SEL) was also examined. Paired-samples t-tests indicated statistically significant improvements within the experimental group in overall SEL scores from pre to post (SEL Total: $t(72) = -2.19$, $p = .032$, Cohen's d = 0.26) and in the Social Awareness domain ($t(72) = -2.13$, $p = .037$, Cohen's d = 0.25), both reflecting small effect sizes. In contrast, no significant changes were observed in the control group on any SEL domain. Furthermore, no significant differences were found between groups at pretest (SEL Total: $t(99) = -0.50$, $p = 0.620$), confirming baseline equivalence and supporting the validity of the comparison.

Further analysis at the group level revealed that the SEL improvement was most pronounced in Group A, where a statistically significant increase in overall SEL scores from pre to post was observed ($t(20) = -2.30$, $p = 0.032$, Cohen's d = 0.50), reflecting a medium effect size. All five individual SEL domains in Group A showed positive pre post change. In contrast, Groups B and C did not show statistically significant changes in any domain.

It is important to note that Groups A, B, and C differed only in the simulation scenario assigned to them, while the underlying Socratic dialogue mechanism and AI model remained identical across all three conditions. The stronger gains observed in Group A may therefore reflect a scenario-specific effect. However, although Group A exhibited the strongest improvement trend, the one way ANOVA did not reveal statistically significant differences between Groups A, B, and C.**The full statistical results are presented in Tables 9 through 12.**

All statistical analyses reported in this section were conducted using a Google Colab notebook, publicly available at [LINK].

Table 9: User Experience (UEQ-S) Comparison Between Experimental and Control Groups

UEQ Dimension	Experimental Mean	Control Mean	t	**p**	Cohen's d	Effect Size
Pragmatic Quality	2.195	1.452	2.510	***0.014**	0.573	Medium
Hedonic Quality	2.183	1.173	3.231	****0.002**	0.738	Medium
Overall Score	2.159	1.312	2.963	****0.004**	0.677	Medium

*Note. Scale: -3 to +3. Experimental N=73, Control N=26. * p <0.05, ** p < 0.01. All three UEQ dimensions reached statistical significance with medium effect sizes. Cohen's d represents the standardized effect size, where 0.2 indicates a small effect, 0.5 a medium effect, and 0.8 a large effect.*

Table 10: Overall SEL Score Comparison Between Experimental and Control Groups

Phase	Experimental Mean	Control Mean	t	p
PRE	3.219	3.254	-0.498	0.6196
POST	3.281	3.248	0.379	0.7052

Note. *No significant differences were found between the experimental and control groups in SEL_Total (all p > 0.05).*

Table 11: SEL Pre → Post Paired Comparisons (Experimental & Control Groups)

SEL Domain	Mean Pre	Mean Post	t	p	Cohen's d	Effect Size
Experimental Group (N=73)						
Self-Awareness	3.298	3.346	-1.146	0.256	0.134	negligible
Self-Management	3.161	3.240	-1.555	0.124	0.182	negligible
Social Awareness	**3.158**	**3.264**	**-2.125**	**0.037**	**0.249**	**small**
Relationship Skills	3.171	3.233	-1.223	0.225	0.143	negligible
Decision Making	3.301	3.322	-0.435	0.665	0.051	negligible
SEL Total	**3.219**	**3.281**	**-2.192**	**0.032**	**0.257**	**small**
Control Group (N=26)						
Self-Awareness	3.288	3.298	-0.189	0.852	0.037	negligible
Self-Management	3.183	3.260	-1.000	0.327	0.196	negligible
Social Awareness	3.192	3.135	0.592	0.559	-0.116	negligible
Relationship Skills	3.365	3.298	1.098	0.283	-0.215	small
Decision Making	3.298	3.250	0.817	0.422	-0.160	negligible
SEL Total	3.265	3.248	0.424	0.675	-0.083	negligible

*Note. Paired samples t tests. Experimental N=73, Control N=26. * p < .05. Only the experimental group showed significant improvement in Social Awareness (d=0.249) and SEL Total (d=0.257). No significant changes were observed in the control group (all p > .28).*

Table 12: Overall SEL Score Pre-Post Comparison by Group

Group	Mean Pre	Mean Post	t	p	Cohen's d	Effect Size
Group A	**3.157**	**3.307**	**-2.301**	**0.0323**	**0.502**	**Medium**
Group B	3.248	3.264	-0.411	0.6846	-	-
Group C	3.237	3.276	-0.864	0.3954	-	-
Group D	3.265	3.248	-2.125	0.6754	-	-

6.1.2 First Year Students: User Experience (UEQ-S) and SEL Outcomes: Experimental vs. Control

The UEQ-S results (Table 13) for first-year students revealed a consistent pattern of higher ratings among experimental participants compared to the control group across all three dimensions, with medium effect sizes throughout (Cohen's d = 0.52–0.59). However, given that Shapiro-Wilk tests indicated violations of normality in the experimental group across all UEQ dimensions (p < .05), Mann-Whitney U tests were conducted as non-parametric alternatives. These non-parametric tests confirmed statistically significant differences across all three dimensions: Pragmatic Quality (U = 492.5, p = .012), Hedonic Quality (U = 486.5, p = .019), and Overall Score (U = 485.0, p = .022). The parametric t-tests approached but did not reach conventional significance thresholds (all p < .10). Taken together, these findings suggest that first year students in the experimental group perceived the system as more usable and more engaging than those in the control group, consistent with the pattern observed among advanced students.

As with advanced students, the most likely explanation for this difference is the Socratic dialogue component. While control participants received only a static CASEL analysis, experimental participants engaged in an interactive, personalized AI-guided conversation that appears to have transformed the experience into an active reflective interaction. This is reflected particularly in the Hedonic Quality dimension (Cohen's d = 0.52), which captures novelty and stimulation (see Table 13).

Table 13: User Experience (UEQ-S) Comparison Between Experimental and Control Groups for First Year Students

UEQ Dimension	Experimental Mean	Control Mean	t	p	Cohen's d	Effect Size	**Mann-Whitney p**
Pragmatic Quality	2.420	1.781	1.901	0.069	0.588	Medium	***0.012**
Hedonic Quality	2.142	1.469	1.835	0.077	0.522	Medium	***0.019**
Overall Score	2.281	1.625	1.971	0.059	0.582	Medium	***0.022**

Note. Scale: −3 to +3. Experimental N=44, Control N=16. The parametric t-test did not reach conventional significance (all p > .05). Given normality violations (Shapiro-Wilk p < .05 for the experimental group), Mann-Whitney U tests were applied as non-parametric alternatives and revealed significant differences across all three dimensions (all p < .05), with medium effect sizes. Cohen's d represents the standardized effect size, where 0.2 = small, 0.5 = medium, 0.8 = large.

Beyond user experience, the impact of the intervention on SEL was examined. In contrast to advanced students, where improvements were modest (small effect sizes), first year students in the experimental group demonstrated substantially stronger and more consistent SEL gains across all five CASEL domains. Paired-samples t-tests revealed highly significant improvements from pre to post across all domains, with large to medium effect sizes. No significant changes were observed in the control group on any domain, and all control group effect sizes were negligible. These findings strongly support the effectiveness of the Socratic dialogue intervention for first-year students (see Table 14).

Table 14: SEL Pre → Post Paired Comparisons for Experimental and Control Groups in First Year Students

SEL Domain	Mean Pre	Mean Post	t	p	Cohen's d	Effect Size
Experimental Group (N=47)						
Self-Awareness	**3.218**	**3.638**	**−5.551**	*****<.001**	**0.810**	**Large**
Self-Management	**3.160**	**3.543**	**−5.118**	*****<.001**	**0.747**	**Medium**
Social Awareness	**3.074**	**3.511**	**−5.612**	*****<.001**	**0.819**	**Large**
Relationship Skills	**3.144**	**3.596**	**−5.326**	*****<.001**	**0.777**	**Medium**
Decision Making	**3.229**	**3.633**	**−5.126**	*****<.001**	**0.748**	**Medium**
SEL Total	**3.165**	**3.281**	**−6.414**	*****<.001**	**0.936**	**Large**
Control Group (N=16)						
Self-Awareness	3.391	3.406	−0.235	.817	0.059	negligible
Self-Management	3.281	3.266	0.187	.855	−0.047	negligible
Social Awareness	3.234	3.297	−0.577	.572	0.144	negligible
Relationship Skills	3.344	3.344	0.000	1.000	0.000	Negligible
Decision Making	3.438	3.438	0.000	1.000	0.000	negligible
SEL Total	3.338	3.350	−0.355	.728	0.089	negligible

*Note. Paired samples t-tests. Experimental N=47, Control N=16. *** p < .001. The experimental group showed highly significant improvements across all five SEL domains and in overall SEL Total, with large to medium effect sizes. The control group showed no significant change in any domain (all p > .57), with negligible effect sizes throughout.*

Further analysis at the group level revealed that all three experimental groups (A, B, and C) demonstrated significant pre-to-post SEL improvements, though the pattern and magnitude of gains varied across groups. Group A showed the strongest and most consistent gains across all domains, with large effect sizes throughout (Cohen's d = 0.894 –1.453). Group B showed significant improvements in four of five domains, with medium to large effect sizes. Group C showed significant improvements in three of five domains, with medium effect sizes. In contrast, Group D (control) showed no significant changes on any domain, with all effect sizes negligible. These results suggest that the Socratic dialogue intervention produced robust and meaningful SEL gains among first-year students, with Group A exhibiting particularly pronounced effects. As with advanced students, Groups A, B, and C differed only in the simulation scenario assigned to them, while the Socratic dialogue mechanism remained identical. The differential gains across groups may therefore reflect scenario specific effects on engagement and depth of reflection(see Table 15).

Table 15: Overall SEL Score Pre - Post Comparison by Group for First Year Students

Group	Mean Pre	Mean Post	t	p	Cohen's d	Effect Size
Group A	**3.097**	**3.618**	**−5.177**	*****<.001**	**1.256**	**Large**
Group B	**3.107**	**3.523**	**−3.285**	****0.005**	**0.848**	**Large**
Group C	**3.300**	**3.607**	**−2.685**	***0.018**	**0.693**	**Medium**
Group D	3.338	3.350	−0.355	.728	0.089	Negligible

*Note. Paired-samples t-tests by group. Group A N=17, Group B N=15, Group C N=15, Group D N=16. * p < .05, ** p < .01, *** p < .001. All experimental groups showed statistically significant pre-to-post improvements. Group D (control) showed no significant change.*

To examine whether SEL gains differed significantly across all four groups, a one way ANOVA on SEL change scores was conducted. Results revealed significant between group differences in SEL Total change ($F = 5.018$, $p = .004$), Self-Awareness change ($F = 4.945$, $p = .004$), Relationship Skills change ($F = 4.052$, $p = .011$), Decision Making change ($F = 3.491$, $p = .021$), and Self-Management change ($F = 2.817$, $p = .047$). Among the three experimental groups only (ANOVA A/B/C), no significant differences were found on any domain, indicating that while all experimental groups outperformed the control, the differences among experimental groups themselves were not statistically significant.

6.1.3 Comparison Between First Year and Advanced Students

To further examine whether the effectiveness of the intervention differed by academic experience level, an additional comparison was conducted between first year students and advanced students. As shown in **Tables 16–18**, Baseline analyses revealed no significant differences between the cohorts on any SEL domain prior to the intervention. However, following the intervention, first year students demonstrated significantly higher post-intervention SEL scores and significantly larger SEL gains across all five CASEL domains and in overall SEL. The largest difference was observed for SEL Total change, where first-year students showed substantially greater improvement than advanced students. In contrast, no significant differences were found between the cohorts on any UEQ-S dimension, indicating that both groups perceived the system similarly despite the stronger SEL gains among first year students. These findings suggest that the Socratic dialogue intervention may be particularly beneficial for students at the beginning of their academic studies **(see Tables 16–18)**.

Table 16: Overall SEL Scores Comparison Between First Year and Advanced Students

SEL Domain	First Year Mean	Advanced Mean	t	p	Cohen's d	Effect Size
PRE	3.216	3.228	-0.251	.802	-0.039	Negligible
POST	**3.525**	**3.272**	**4.073**	**<.001**	**0.662**	**Medium**

Table 17: SEL Change Scores Comparison Between First Year and Advanced Students

SEL Domain	First Year Mean	Advanced Mean	t	p	Cohen's d	Effect Size
Self-Awareness	**0.317**	**0.038**	**3.927**	**<.001**	**0.689**	**Medium**
Self-Management	**0.282**	**0.078**	**2.671**	**.009**	**0.448**	**Small**
Social Awareness	**0.341**	**0.063**	**3.442**	**<.001**	**0.576**	**Medium**
Relationship Skills	**0.337**	**0.028**	**3.802**	**<.001**	**0.657**	**Medium**
Decision Making	**0.302**	**0.003**	**3.925**	**<.001**	**0.679**	**Medium**
SEL Total	**0.316**	**0.042**	**4.619**	**<.001**	**0.839**	**Large**

Table 18: One Way ANOVA Results for SEL Change Scores Between First Year and Advanced Students

SEL Domain	F	p
Self-Awareness Change	**19.318**	**<.001**
Self-Management Change	**7.985**	**.005**
Social Awareness Change	**13.184**	**<.001**
Relationship Skills Change	**17.544**	**<.001**
Decision Making Change	**18.519**	**<.001**
SEL Total Change	**30.119**	**<.001**

Note. Results represent the main effect of Cohort (First Year vs. Advanced Students). Significant differences indicate that first year students demonstrated larger SEL gains than advanced students.

6.1.4 Overall UEQ-S Comparison: Experimental vs. Control Groups

To provide a comprehensive overview of user experience outcomes across both cohorts, an additional analysis was conducted combining the UEQ-S scores of Advanced Students and First Year Students into a single cross-cohort comparison. Averaged across both cohorts, experimental participants consistently rated the system higher than control participants across all three UEQ-S dimensions: Pragmatic Quality (Experimental M = 2.31, Control M = 1.62), Hedonic Quality (Experimental M = 2.16, Control M = 1.32), and Overall Score (Experimental M = 2.22, Control M = 1.47). All differences were statistically significant with medium effect sizes, as reported in Sections 6.1.1 and 6.1.2. **Figure 2** illustrates this pattern visually.

The consistency of this pattern across both cohorts, despite differences in academic experience, recruitment context, and SEL gain magnitude, suggests that the Socratic dialogue component was the primary driver of the enhanced user experience. Whereas control participants received only a static AI generated CASEL analysis, experimental participants engaged in an interactive, personalized reflective conversation that appears to have transformed the platform from a passive assessment tool into an engaging and meaningful learning experience. This interpretation is particularly supported by the Hedonic Quality dimension, which captures novelty, stimulation, and engagement, and which showed the largest gap between conditions. Taken together, these findings indicate that the conversational component not only improved SEL outcomes but also substantially enhanced how participants experienced and perceived the system as a whole **(see Figure 2)**.

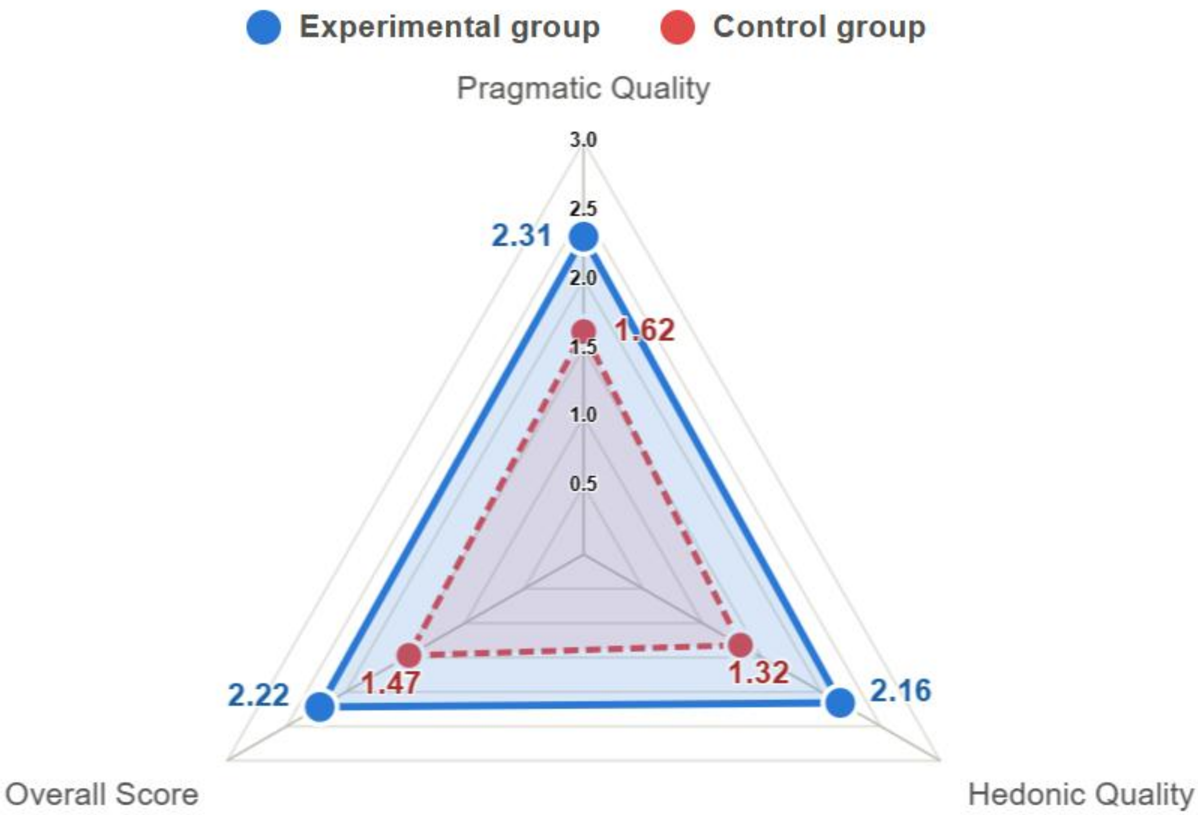

Figure 2. Radar chart of UEQ-S scores across three dimensions: Experimental vs. Control groups (combined cohorts, N = 174).

7. Qualitative analysis of the conversation

Beyond the quantitative outcomes presented above, a qualitative analysis was conducted on the content of the Socratic dialogue and the reflection responses collected from participants in the experimental groups (A, B, C), who were the only participants to engage in the interactive dialogue stage with the Socratic coach. These data provide a deeper understanding of how participants experienced and navigated the reflective process. The data originate from the social emotional simulation completed by each participant, whose response was automatically analyzed across the five CASEL domains (self awareness, self management, social awareness, relationship skills, and responsible decision making), as well as from the subsequent Socratic dialogue, which was grounded in this analysis and extended it toward deeper personal reflection. All data, including simulation responses, CASEL analyses, dialogue transcripts, and reflection answers, were systematically documented and retained for each participant throughout the experiment, forming the basis for the thematic analysis presented below.

The analysis was conducted in two stages. First, broad categories were identified, reflecting recurring cognitive and emotional patterns among participants. Second, each category was further refined into more specific subcategories, with representative quotations and their frequency of occurrence (N) documented. This process was carried out separately for the advanced student cohort and the first year student cohort, in order to examine potential differences between the two populations in how they experienced and interpreted the Socratic dialogue.

7.1 Thematic Findings: Advanced Students

Among advanced students, the dominant themes that emerged reflected the development of key Social Emotional Learning (SEL) competencies aligned with the CASEL framework. Participants described developing greater self awareness by recognizing their own habitual "autopilot" responses under pressure, and exercising self management by learning to pause and separate emotional reactivity from factual reality. Several reflections also highlighted relationship skills and social awareness, particularly in navigating the tension between social loyalty, boundary setting, and academic integrity, wh-ile a recurring shift toward responsible decision making was evident in participants' reframing of help seeking as an act of strength and responsibility rather than weakness. Taken together, these themes illustrate how the Socratic dialogue supported reflective development across the core CASEL competencies. **Table 19** presents the complete set of categories, subcategories, representative quotations, and their frequency of occurrence.

Table 19: Qualitative Themes, Subcategories, Representative Quotes, and Frequencies for Advanced Students

Category	Subcategory	Representative Quote	N
Exiting "Autopilot" and Developing Self Awareness	The "Pause" Technique and Mental Stop	"When we're under pressure and emotionally overwhelmed, our minds tend to switch into autopilot. In those moments, we react from instinct and survival mode rather than from rational thinking. On top of that, stopping a 'runaway train' of negative emotions requires a great deal of self-awareness and mental effort. It's much easier to keep rolling downhill with the frustration than to suddenly hit the brakes and take responsibility."	~35–45
	Separating Facts (Reality) from Emotions (Fear/Stress)	"She could have stopped for a moment and asked herself, 'Do I actually have evidence that Liam is laughing at me, or is this just fear because of how I'm feeling today?' If she had realized that her thought was based on emotion because she was having a bad day rather than on facts, it would have been easier for her to recognize that it was her fear speaking, not reality."	~30–40
The Effect of Pressure, Fear of Failure, and Judgment on Decision Making	Anxiety about Judgment and Fear of Appearing "Weak"	"I think what holds us back the most is the fear that the moment we say, 'I'm struggling,' the entire tower we've built will collapse. We believe that if we admit to a weakness, everyone will suddenly see that we're not as strong or as smart as they thought we were."	~25–35
	Rationalization and Seeking "Shortcuts" (Survival Mode)	"The key issue is that Yotam fell into a pattern of self justification. He kept telling himself stories like, 'It's only bonus credit,' or 'The lecturer is just making us do unnecessary work,' simply to silence his conscience. He was so exhausted and worried about his grade that he chose to close his eyes and push away the feeling that something about what he was was doing was not right."	~30–40
The Conflict Between Social Loyalty, Boundary-Setting, and Academic Integrity	People-Pleasing and Social Pressure	"The problem is that you feel like you can't refuse to help your friend, even if it means breaking the rules."	~45–55
	Redefining Friendship and Setting Boundaries	"I think a friend who truly supports you will also tell you 'no' when necessary because they want what is best for you. A friend who	~25–35

Category	Subcategory	Representative Quote	N
		always agrees is simply going along with you, even when it is not the right thing for you."	
Perceptual Shift: Asking for Help as Strength, Not Weakness	Seeking Help as a Threat to the Ego / Impostor Syndrome	"When I feel under pressure, this realization reminds me that I don't have to face everything on my own. It reminds me that my worth does not depend on being perfect, and that collaboration and openness can lead to better decisions while also reducing my stress."	~20–30
	Asking Questions as an Expression of Responsibility and Confidence	"Asking questions is actually a sign that I want to understand and do things the right way. It helps prevent mistakes and shows responsibility rather than weakness. It's simply part of genuine learning."	~20–25
Transition from Survival Response to Value-Driven Action with Courage	Building Courage and Confidence Through Action (Action Precedes Confidence)	"At first, you feel pressure and discomfort, but afterward comes relief, pride, and self confidence."	~15–25
	Transition from Reactivity to Conscious (Value-Driven) Choice	"Pause for a moment, separate fear from reality, and choose consciously rather than reacting automatically."	~25–35

Note. All representative quotations were translated from Hebrew into English by the authors.

7.2 Thematic Findings: First Year Students

Among first year students, the dominant themes similarly reflected engagement with the core CASEL competencies, though with a stronger emphasis on immediate emotional regulation and the gap between values and real time behavior. Participants described heightened self awareness through recognizing how stress, vulnerability, and fear distort judgment in the moment, and demonstrated self management by discovering the value of pausing before reacting and shifting from outcome focused to process focused thinking. Social awareness and relationship skills emerged in reflections on social loyalty, peer pressure, and the tendency to project one's emotional state onto social situations, while responsible decision making was evident in participants' recognition that stress can drive shortcuts and moral compromise, alongside a notable shift toward viewing help seeking as a sign of strength rather than weakness. These themes illustrate how participants reflected on experiences related to the core CASEL competencies throughout the Socratic dialogue. **Table 20** presents the complete set of categories, subcategories, representative quotations, and their frequency of occurrence.

Table 20. Qualitative Themes, Subcategories, Representative Quotations, and Frequencies for First Year Students

Category	Subcategory	Representative Quote	N
The impact of stress and emotions on decision-making	vulnerability and embarrassment as a factor in avoidance	In the moment of truth, fear becomes stronger than rational thinking.	~25-30
	Rationalization and self-justification in stressful situations	"At the time, it feels completely logical, as if we have a good reason, but only afterward do we realize that we were just convincing ourselves."	~20
Interpersonal dynamics and social pressure	Social loyalty versus integrity and personal values	"Loyalty to yourself is the foundation of loyalty to others. If I stop acting according to my values just to please my friends, the relationship becomes less genuine because I am no longer being true to myself."	~25
	Projection of emotional state onto social reality (cognitive bias)	"When we are in a negative emotional state, fear can sometimes have a stronger influence than what is actually happening, leading us to interpret situations less accurately."	~15-20
Development process	The power of pausing ("stop and check")	"Taking a moment to pause and breathe before reacting, even for just a few seconds, can make a big difference and give me time to think instead of reacting automatically."	~40-50
	Shift from outcome-focused thinking to process-focused thinking (taking ownership)	"In the end, success that does not come from your own effort feels less genuine and less fulfilling, and that feeling stays with you long after the exam is over."	~40-50
The impact of stress on moral decision making	Stress as a driver of shortcuts and cheating (academic and social)	"I think this shows that our values are still there, but under pressure they become weaker or get pushed aside. We do not forget what is right and wrong, it simply becomes harder to act according to those values."	~40-50

Category	Subcategory	Representative Quote	N
Conflict between values and reality	The gap between theoretical knowledge and real-time application	"The problem is not that we do not know what is right, but that we do not always give that knowledge space in the moment."	~20-25
Reshaping perceived self-efficacy	Fear of weakness versus viewing help-seeking as strength	"Asking for help does not mean you are incapable. On the contrary, it shows that you want to understand, improve, and become genuinely better."	~20

Note. All representative quotations were translated from Hebrew into English by the authors.

7.3 Comparison of Shared Themes Between First Year and Advanced Students

The qualitative analysis revealed four core themes that consistently emerged across both first year and advanced student cohorts, suggesting that several social emotional challenges accompany academic decision making throughout the university journey. Nevertheless, clear differences were observed in the ways the two groups interpreted and reflected upon these experiences. Both cohorts highlighted the influence of social pressure and the difficulty of refusing a friend's request when it conflicted with personal values and academic integrity. However, first year students tended to describe this dilemma primarily as an immediate desire to avoid disappointing a friend or creating an uncomfortable situation, whereas advanced students interpreted it as a broader conflict involving interpersonal loyalty, personal identity, and academic integrity. Similarly, both groups identified fear of appearing weak or lacking knowledge as a major barrier to seeking help. First year students generally associated this fear with basic academic insecurity, whereas advanced students described it as a burden of excellence and a concern that asking for help might undermine the image of a competent, high achieving student they had established throughout their studies. Rationalization also emerged as a shared theme, with participants in both cohorts recognizing how individuals justify ethically questionable decisions under pressure. However, advanced students demonstrated a more sophisticated understanding of this process, describing it as a conscious strategy for self justification and the suppression of guilt. Finally, both groups emphasized the importance of pausing before acting, although first year students primarily viewed this pause as a strategy for regulating emotions and preventing impulsive reactions, whereas advanced students described it as a deliberate cognitive strategy that facilitates ethical reflection, evaluation of consequences, and value based decision making. Overall, although both cohorts experienced remarkably similar social emotional challenges, advanced students consistently demonstrated deeper and more reflective interpretations of these experiences, while first year students focused more strongly on the immediate emotional aspects of the situations they encountered.

8. Additional observations

Table 21 presents **spontaneous remarks** documented by the researchers during the field experiment sessions. These observations were recorded during data collection and are classified according to their source and context of occurrence.

Table 21: *Spontaneous Participant Observations Collected During Field Experiment*

Quote / Paraphrase	Source	Context
("Wow, it's never ending!" referring to the Socratic bot, and then, upon being told he could stop at any time, "No no, I'll finish it. Only 3 minutes left.")	First year student	Mid dialogue, approximately 5–6 minutes into the Socratic conversation
("Very interesting, I liked the idea, and seeing how your answers change after the simulation is very interesting. I really liked the Socratic bot.")	Advanced student	After completing the full experimental flow
("This is something innovative. This is the first time I have ever seen something like this.")	Advanced student	Immediately after completing the Socratic dialogue
("The experiment is very long, I suggest you shorten it.")	Advanced student	After completing the full experimental flow
("I liked that during the experiment I could always see exactly where I was and what I had left.")	Advanced student	After completing the full experimental flow
("Very interesting.")	First year student	After completing the full experimental flow
("The bot is very nice.")	First year student	During the Socratic dialogue phase, spoken aloud to a nearby peer
("Which AI model did you use? It was very interesting.")	Advanced student	During the Socratic dialogue phase
("The system does not look like it was built by students. It is very impressive.")	First year student	After completing the full experimental flow
("Overall I engaged with things at the surface level rather than with deeper internal analysis. If the goal is to reach more internal layers, I would suggest emphasizing that more explicitly.")	Advanced student	After completing the full experimental flow
("That was really nice, I liked it")	Advanced student	After completing the full experimental flow
("Your research is very interesting.")	Advanced student	After completing the full experimental flow

Note. All remarks were collected during the field experiment sessions. Participant attribution is indicated by academic level (First Year or Advanced) to preserve anonymity. Remarks were translated from Hebrew into English by the authors.

9. Discussion

This study addressed the central research question: How does an AI based Socratic dialogue influence the development of Social Emotional Learning (SEL) skills among higher education students? Overall, the findings indicate that a brief AI based Socratic dialogue can meaningfully support SEL development by promoting a structured reflective process rather than simply providing automated feedback. High completion rates across both cohorts, together with consistently positive user experience ratings, indicate that students were willing to engage with the reflective dialogue despite differences in academic experience and recruitment context. Beyond demonstrating the feasibility of the intervention, the quantitative findings showed significantly greater improvements among experimental participants than among controls, suggesting that the conversational component itself played a central role in facilitating change. These findings are consistent with previous evidence demonstrating that structured SEL interventions can strengthen emotional regulation, interpersonal skills, and the management of academic stress through guided reflection and active engagement (Johnson, 2020). Furthermore, the present findings address the research gap concerning technology mediated approaches to SEL in higher education by providing empirical evidence that an AI based Socratic dialogue can serve as a structured reflective intervention for supporting social emotional development among higher education students, extending the direction for future research proposed in recent literature (Wang & Ishak, 2025).

The task completion findings further support the feasibility of the CASELy platform as both a research and educational tool. Although advanced students demonstrated consistently higher completion rates than first year students, engagement remained high across both cohorts despite differences in academic experience and recruitment context. The slightly lower completion observed among first year students may reflect their limited familiarity with reflective learning activities and AI mediated dialogue rather than reduced acceptance of the intervention. Importantly, participants who completed the Socratic dialogue devoted substantially more time to the intervention than those in the control condition, suggesting that they actively engaged with the reflective conversation rather than simply progressing through the required tasks. Informal feedback collected during the study further reinforced this interpretation, with many participants describing the dialogue as engaging and expressing interest in continuing the conversation. Together, these findings demonstrate that the platform successfully sustained participant engagement while maintaining a brief and practical intervention suitable for higher education settings.

The quantitative findings provide strong evidence that the Socratic dialogue was the primary factor underlying the observed improvements in SEL. Across both cohorts, experimental participants demonstrated significant pre to post improvements, whereas the control groups, which received only the AI generated CASEL analysis, showed no significant changes despite comparable baseline scores. Among advanced students, statistically significant improvements were observed for SEL Total ($p = .032$) and Social Awareness ($p = .037$), indicating that the intervention produced measurable, although relatively modest, gains. In contrast, first year students demonstrated highly significant improvements across all five CASEL competencies and overall SEL (all $p < .001$), suggesting a substantially stronger response to the intervention. These findings suggest that students at the beginning of their academic journey may be more responsive to structured reflective interventions, whereas advanced students, who may already possess more established reflective and coping strategies, exhibited smaller but still meaningful quantitative improvements.Interestingly, among advanced students, only Group A demonstrated a statistically significant improvement in overall SEL ($p = .032$), suggesting that different simulation scenarios may vary in their ability to stimulate reflective thinking and SEL development. The consistently higher UEQ-S ratings reported by experimental participants across both cohorts further support this interpretation, particularly for Hedonic Quality, indicating that the conversational component transformed the platform from a passive assessment into an engaging reflective learning experience. This finding is consistent with previous literature emphasizing that embedding SEL principles into learning design can enhance student engagement, well being, and learning outcomes (DeSensi, 2024). It also complements recent recommendations to further investigate technology mediated SEL interventions in higher education (Wang & Ishak, 2025).

While the quantitative findings demonstrated that the AI based Socratic dialogue improved students' SEL competencies, the qualitative findings provide important insight into how these improvements occurred. Across both cohorts, participants consistently described a shift from automatic, emotionally driven reactions toward more deliberate, reflective, and value aligned decision making. They became increasingly aware of how fear, stress, social pressure, and concerns about being judged shaped their thinking, learning to pause before reacting, distinguish emotions from objective reality, question their assumptions, and consider alternative responses. These patterns closely reflect the CASEL competencies of self awareness, self management, social awareness, and responsible decision making, and are consistent with research suggesting that effective SEL in higher education is strengthened when students actively engage in reflective practice, receive opportunities to practice social emotional skills, and obtain feedback rather than merely receiving information (Conley & Donahue-Keegan, 2024). They also align with evidence that emotional states such as stress and anxiety can shape students' thinking and decision making in academic contexts, highlighting the importance of reflective processes that help students regulate emotions before acting (Simion, 2023).

Participants also challenged patterns of rationalization and self justification, reconsidered conflicts between social loyalty and personal values, and reframed help seeking as an expression of responsibility and maturity rather than weakness. Although similar themes emerged across both cohorts, first year students tended to emphasize immediate emotional experiences, whereas advanced students more often interpreted the same situations in relation to academic identity, ethical responsibility, and longer term consequences. Participants across both cohorts also emphasized the importance of pausing before responding under pressure and recognized that acting in accordance with personal values often precedes feelings of confidence. Importantly, these insights emerged without direct instruction or explicit feedback, suggesting that an AI based Socratic dialogue can support reflective learning through guided self reflection. These findings extend previous higher education research by illustrating how students reconstruct beliefs about responsibility, help seeking, and value based decision making through the reflective process, complementing recent work highlighting the importance of strengthening SEL competencies and expanding SEL implementation within higher education (Ahmad & Singh, 2025; Ren et al., 2025).

10. Contributions

10.1 Theoretical contribution

This study contributes to the theoretical understanding of Social Emotional Learning (SEL) in higher education by addressing a gap identified in recent literature regarding AI mediated approaches to SEL (Wang & Ishak, 2025). The findings provide empirical evidence that a brief AI guided Socratic dialogue supported the development of SEL competencies through structured self reflection rather than direct instruction. While previous research has established the value of SEL interventions in higher education, little is known about how AI based Socratic dialogue may facilitate this process. The present findings extend this literature by demonstrating that an anonymous, instructor independent dialogue produced measurable improvements in SEL while also promoting meaningful reflective processes across both student cohorts. Furthermore, the comparison between first year and advanced students suggests that academic experience may influence responsiveness to AI guided reflective interventions, with stronger improvements observed among students at the beginning of their academic studies. Finally, the qualitative findings provide insight into a possible mechanism underlying these changes, showing that participants gradually shifted from automatic, emotionally driven reactions toward more deliberate, value aligned decision making through their own reflective reasoning rather than through explicit instruction or evaluative feedback.

10.2 Practical Contribution

The central practical contribution of this project is the creation of a robust research infrastructure (CASELy), which serves as an operational and sustainable tool enabling researchers to conduct SEL studies in a controlled, reliable, and standardized manner. The system was designed with an emphasis on flexibility and scalability. Owing to its linear architecture, the platform ensures consistent data collection and minimizes research bias, such as skipping stages or altering the sequence of tasks, while also automatically documenting engagement data and completion times for future statistical analyses.

11. Limitations

A primary limitation of this study concerns participant recruitment. Although the study included students from nine academic institutions, the majority of participants (146 of 174; 83.9%) were recruited from Braude College of Engineering, which may limit the generalizability of the findings to other higher education contexts. In addition, participation was entirely voluntary across both cohorts, raising the possibility of self selection bias, as students who chose to participate may have been more willing to engage in reflective learning activities or AI based interventions than those who did not participate. Nevertheless, the sample included students from multiple academic institutions, diverse fields of study, a broad age range (18–36+), both genders, and two distinct academic cohorts (first year and advanced students), providing meaningful diversity despite the voluntary recruitment process.

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Appendix A: Research Instrument from Ahmad & Singh (2025)

Construct	Items
Self-Awareness	[SA1] I am aware of the emotions I feel. [SA2] I can calm myself down. [SA3] I know what my strengths are. [SA4] I am aware when my feelings are making it hard to focus.
Self-Management	[SM1] I can be patient during lessons that get me excited. [SM2] I always manage to finish my tasks even if they are hard for me. [SM3] I can set goals for myself. [SM4] I complete my assignment even if I do not feel like it.
Social-Awareness	[SoA1] I learn from people with different opinions than me. [SoA2] I am able to tell what people may be feeling. [SoA3] I know when someone needs help. [SoA4] I know how to get help when I'm having trouble with a classmate.
Relationship-Skills	[RS1] I can respect a classmate's opinions during disagreements. [RS2] I get along well with my classmates. [RS3] I always speak to an adult when I have problems at school. [RS4] I build and maintain healthy relationships within my university community.
Responsible-Decision-Making	[RDM1] I think about what might happen before making any decision. [RDM2] In a situation, I know what is right or wrong. [RDM3] I can strictly say "NO" to a friend who wants me to break the rules. [RDM4] I always seek advice or feedback from others before making important decisions.

Appendix B: Full UEQ-S Questionnaire

obstructive	o o o o o o o	supportive	1
complicated	o o o o o o o	easy	2
inefficient	o o o o o o o	efficient	3
clear	o o o o o o o	confusing	4
boring	o o o o o o o	exciting	5
not interesting	o o o o o o o	interesting	6
conventional	o o o o o o o	inventive	7
usual	o o o o o o o	leading edge	8

Appendix C: User Guide

1. Overview

EduMap is an AI powered web platform based on the CASEL framework, designed to strengthen the five core Social Emotional Learning competencies while providing intelligent support for both students and educators. The platform consists of three main components: the Research Participation Module, the Student Portal, and the Educator Portal. As part of this capstone project and the accompanying research on Social Emotional Learning in higher education, a dedicated Research Participation Module was developed and became the central component of the system. This module integrates validated assessments, AI generated simulations, and CASELy, an AI guided Socratic coach that promotes reflective learning through Socratic dialogue. It supports the complete research workflow, including questionnaires, simulations, AI based analysis, reflective conversations, and user experience evaluation.In addition to the Research Participation Module, the platform also includes dedicated Student and Educator portals that support learning and instructional activities. However, since the primary focus of this capstone project is the research process, this user guide focuses exclusively on the Research Participation Module and the participant workflow. An overview of the platform is shown in **Figure 1.**

Additional Resources and Demonstration Videos

The following links provide access to the EduMap platform and demonstration videos illustrating the main components presented in this guide.
System Website: https://edumapcasely.online/
Research Participation Module Demo: https://www.youtube.com/watch?v=KxAtBzIypjE
Educator Portal Demo: https://www.youtube.com/watch?v=Z8i5PYHrlVU&t=16s
Student Portal Demo: https://www.youtube.com/watch?v=12pFgtdFHpM

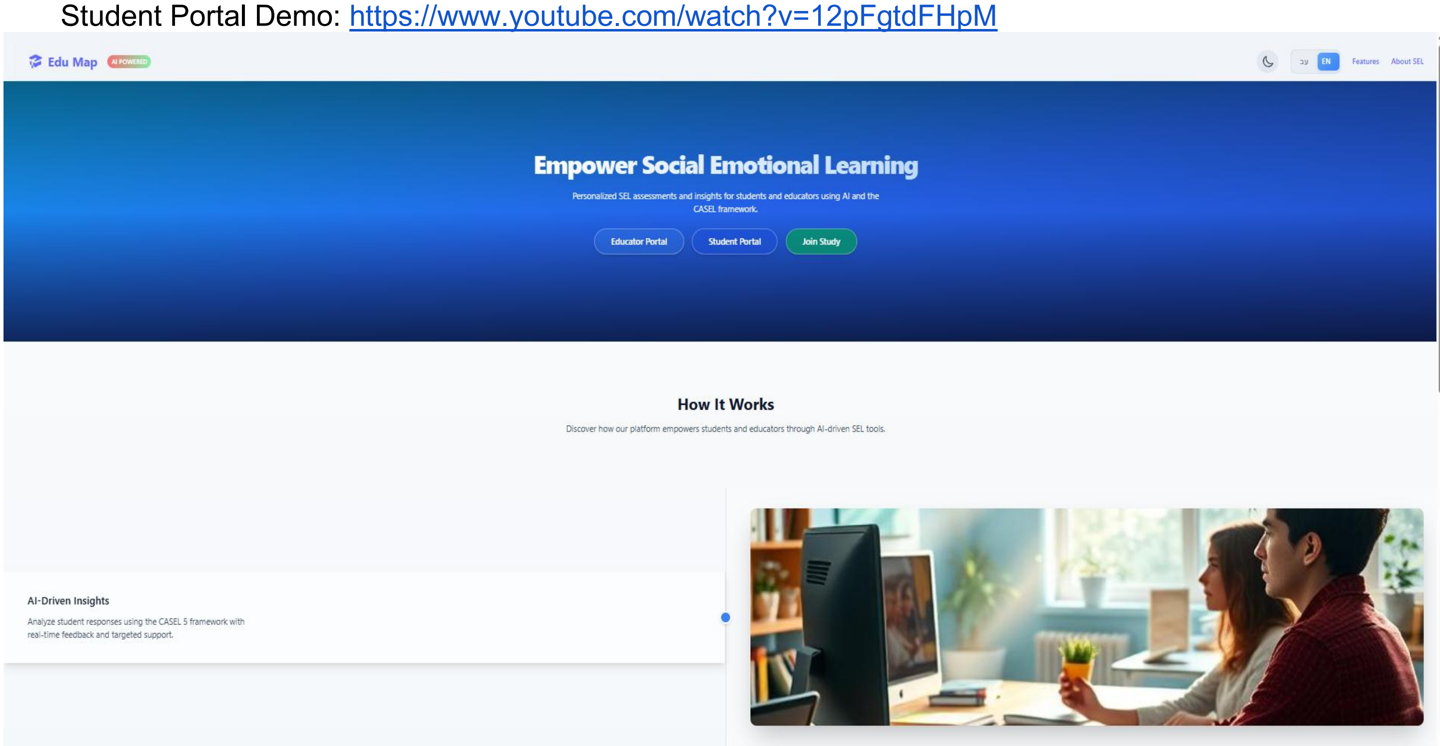

Figure 1. Overview of the EduMap platform home page and its main modules

2. Research Participation Module

The research process begins from the EduMap home page shown in **Figure 1**. Selecting the Join Study button opens the research entry page shown in **Figure 2**.

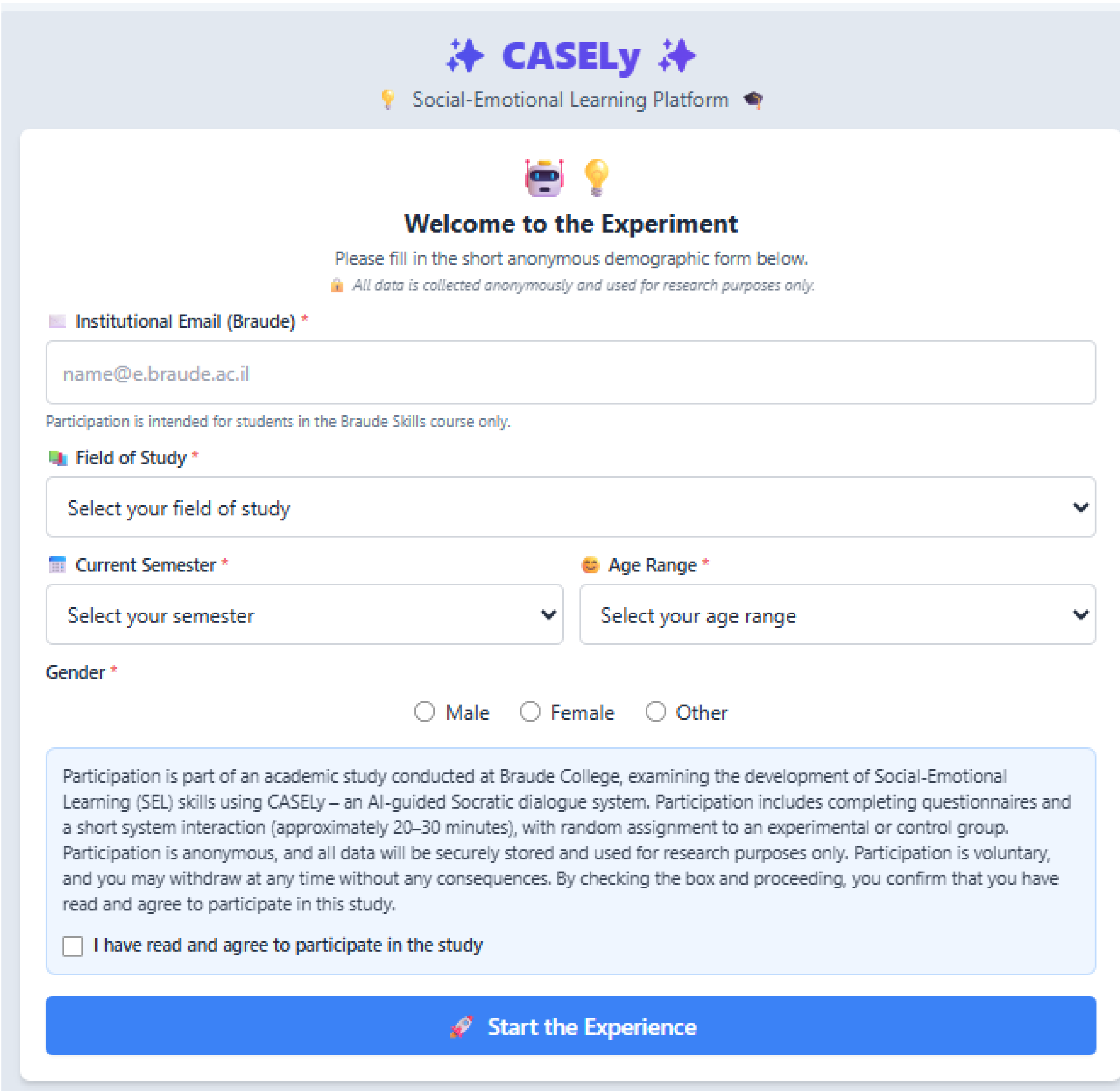

Figure 2. Research entry page and demographic form

The research entry page serves as the starting point of the experimental workflow. Participants provide the required demographic information, including field of study, current semester, age range, and gender. The page also presents information about the study and the consent statement. After confirming participation by selecting the consent checkbox, the Start the

Experience button becomes available. Selecting this button initiates the research workflow and transfers the participant to the next stage.

After selecting the **Start the Experience** button, participants are presented with a page that provides an overview of the stages they will complete during the study. The sequence displayed depends on the group to which the participant was assigned. Participants in the experimental group are shown a seven step journey, including the pre intervention SEL questionnaire, simulation scenario, AI supported Socratic dialogue, brief summary, post intervention questionnaire, final reflections, and the user experience questionnaire. This page helps participants understand the activities they will complete before continuing to the next stage. **Figure 3** illustrates the overview presented to participants in the experimental group.

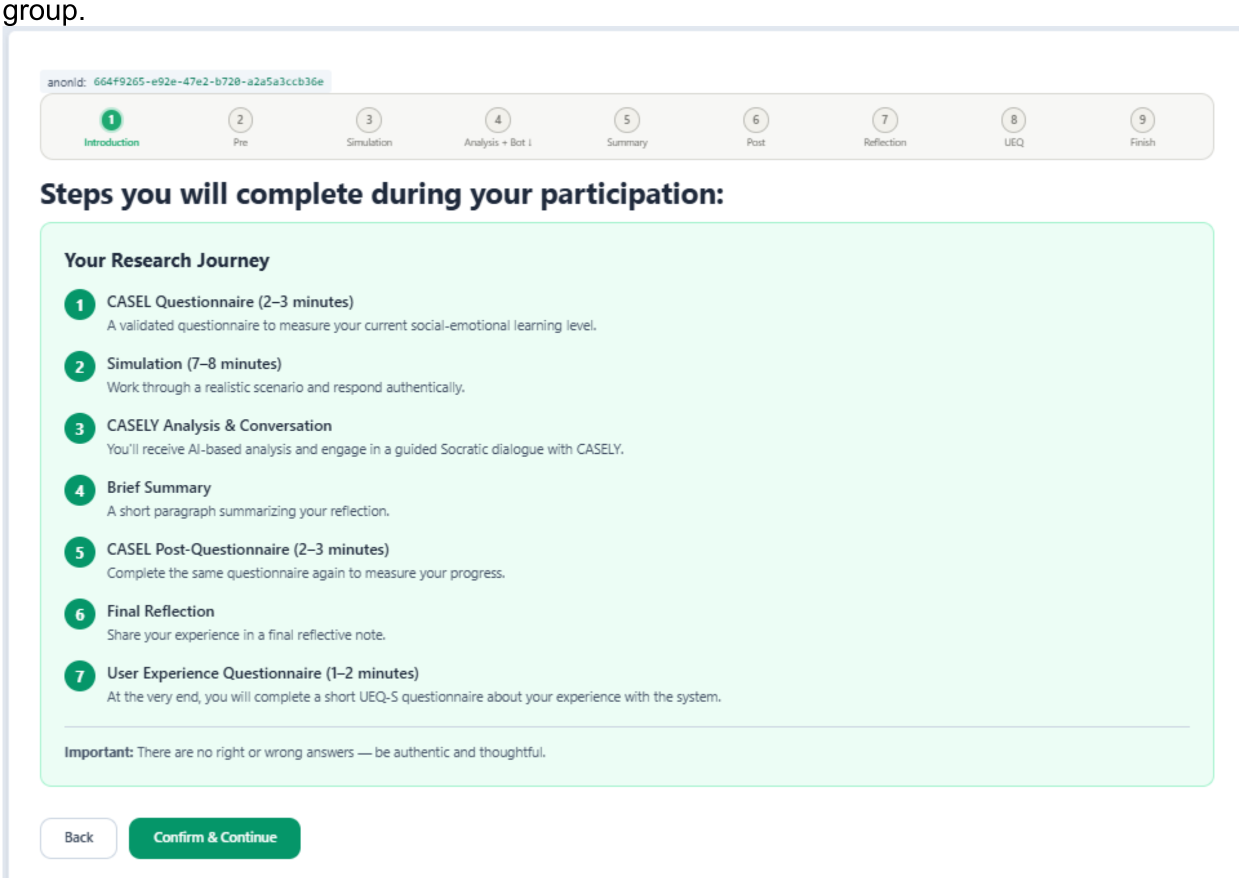

Figure 3. Experimental group research journey

Participants assigned to the control group are presented with a simplified five step journey. This sequence includes the pre intervention SEL questionnaire, simulation scenario, AI generated feedback without an interactive dialogue, post intervention questionnaire, and the user experience questionnaire. Similar to the experimental group, this page provides participants with an overview of the activities they will complete before proceeding to the next stage. **Figure 4** illustrates the overview presented to participants in the control group.

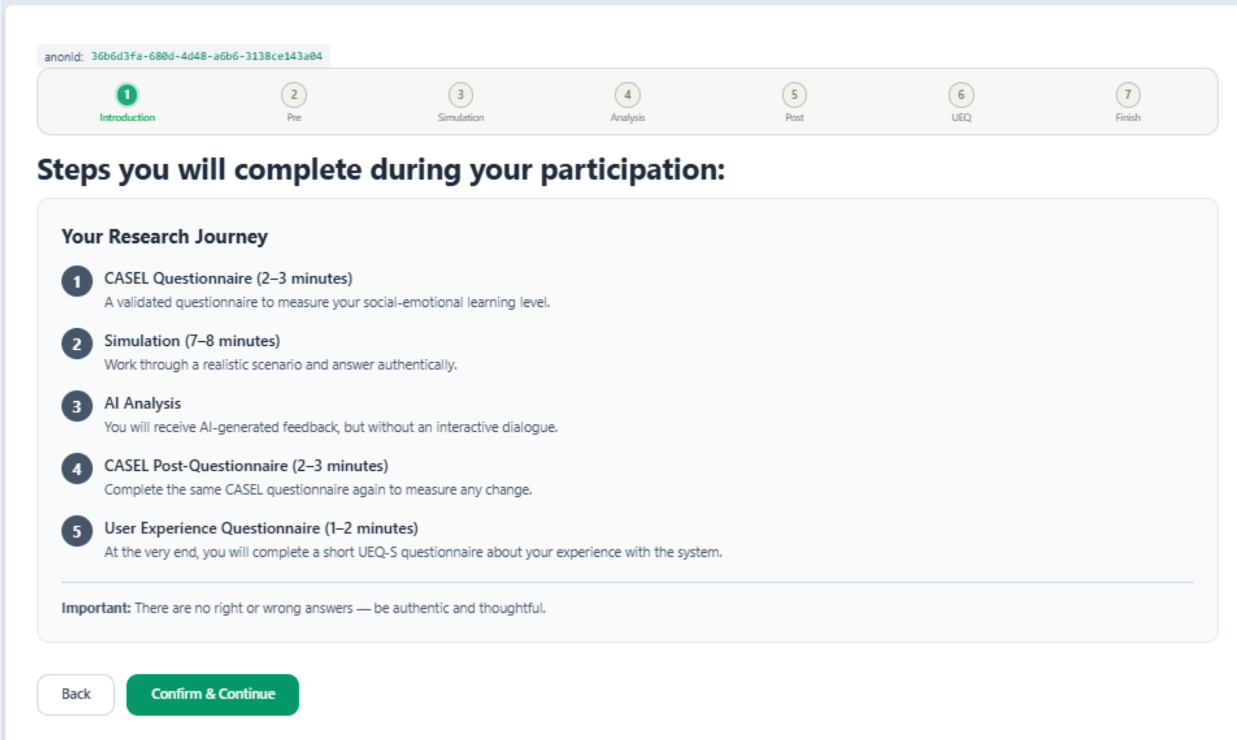

Figure 4. Control group research journey

After selecting **Confirm & Continue,** participants are transferred to the SEL Skills Questionnaire page. This page introduces the five CASEL competencies assessed by the questionnaire and provides information about the estimated completion time. Participants may begin the assessment by selecting the Begin Questionnaire button. **Figure 5** illustrates the SEL Skills Questionnaire introduction page.

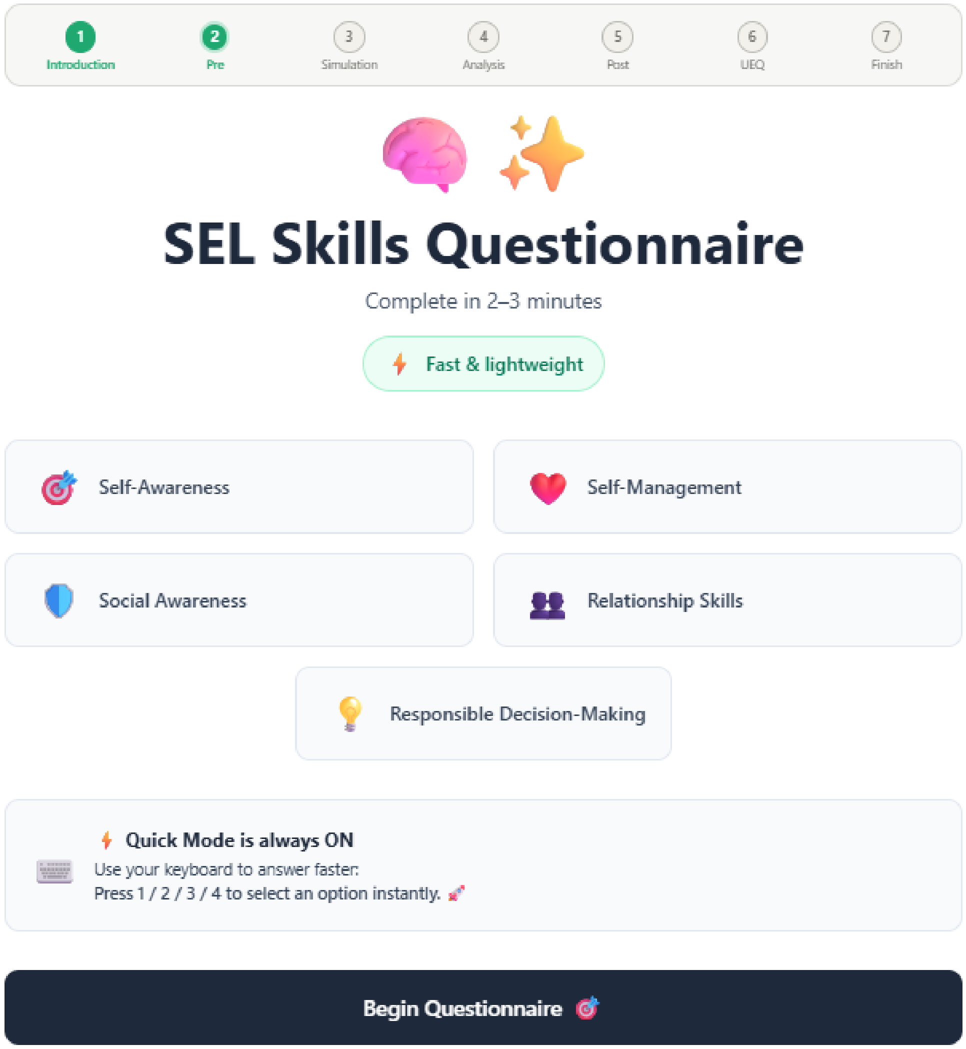

Figure 5. SEL Skills Questionnaire introduction page

After selecting **Begin Questionnaire**, participants proceed to the questionnaire itself. The assessment consists of 20 statements covering the five CASEL competencies. For each statement, participants select one of four response options ranging from Strongly Disagree to Strongly Agree. A progress indicator displayed at the top of the page informs participants about their current progress throughout the questionnaire. **Figure 6** illustrates an example of a questionnaire item.

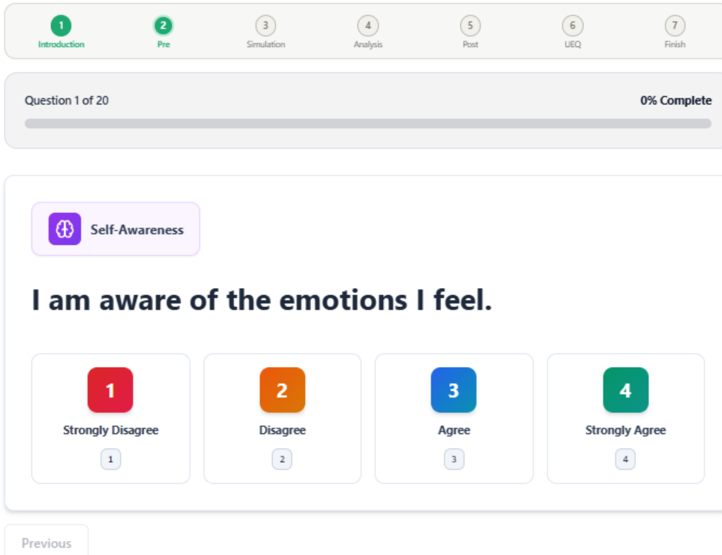

Figure 6. SEL Skills Questionnaire item

After completing all questionnaire items, a confirmation message is displayed automatically, indicating that the questionnaire has been completed successfully. To continue, participants should select the **View Results** button. **Figure 7** shows the completion message displayed after the questionnaire is finished.

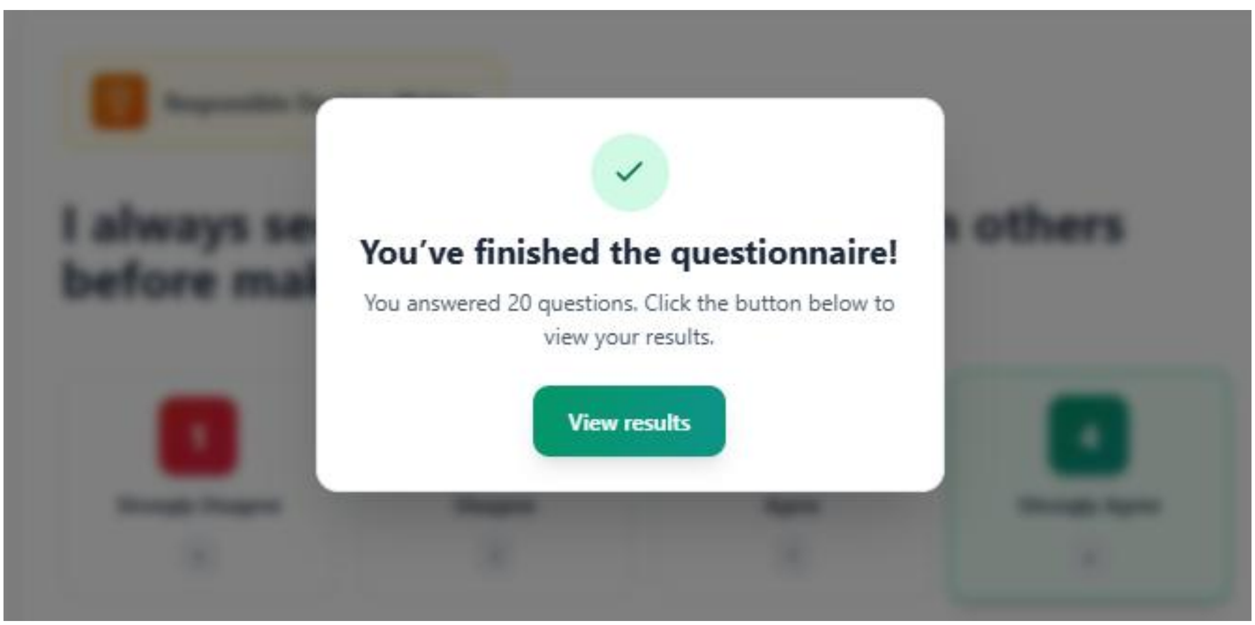

Figure 7. Questionnaire completion message

After selecting the **View Results** button, participants are presented with a summary page displaying their scores across the five SEL competencies. The page provides a visual overview of the results and highlights individual strengths. To proceed to the next stage, participants should select the **Finish** button. **Figure 8** shows the questionnaire results page.

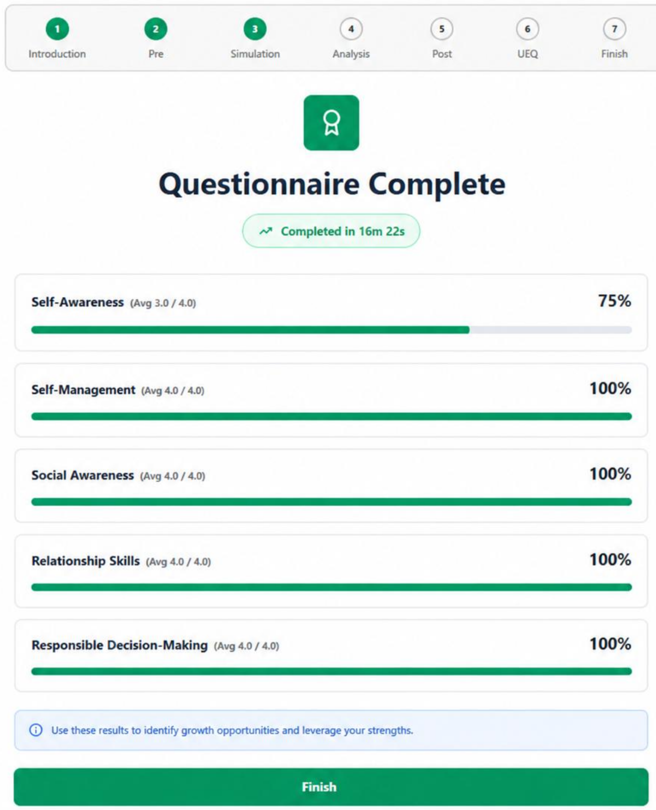

Figure 8. Pre intervention questionnaire results

After selecting the **Finish** button on the questionnaire results page, participants are automatically transferred to the simulation stage. The simulation page presents a scenario and one or more questions related to the situation. Participants are asked to read the scenario carefully and provide their responses in the designated fields. A timer displayed at the top of the page indicates the remaining time available for completing this stage. **Figure 9** shows an example of the simulation page.

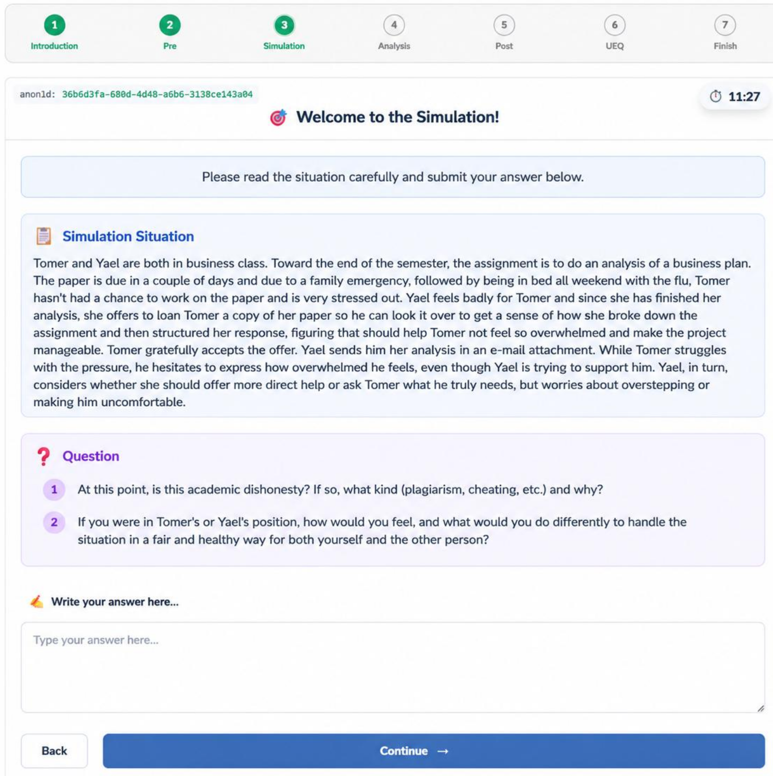

Figure 9. Simulation page

After reviewing the scenario and entering their responses, participants may proceed by selecting the **Continue** button. This action transfers them to the analysis stage, where AI generated feedback based on their responses is displayed. The analysis page presents the participant's submitted answer together with a CASEL based analysis across the five social emotional competencies. For each competency, the page provides a score and a brief explanation describing the strengths reflected in the participant's response. In addition, the page highlights observed strengths, areas for improvement, and a suggested intervention aimed at supporting further development.

Participants are encouraged to review the feedback carefully before proceeding to the next stage. **Figure 10** illustrates the analysis page.

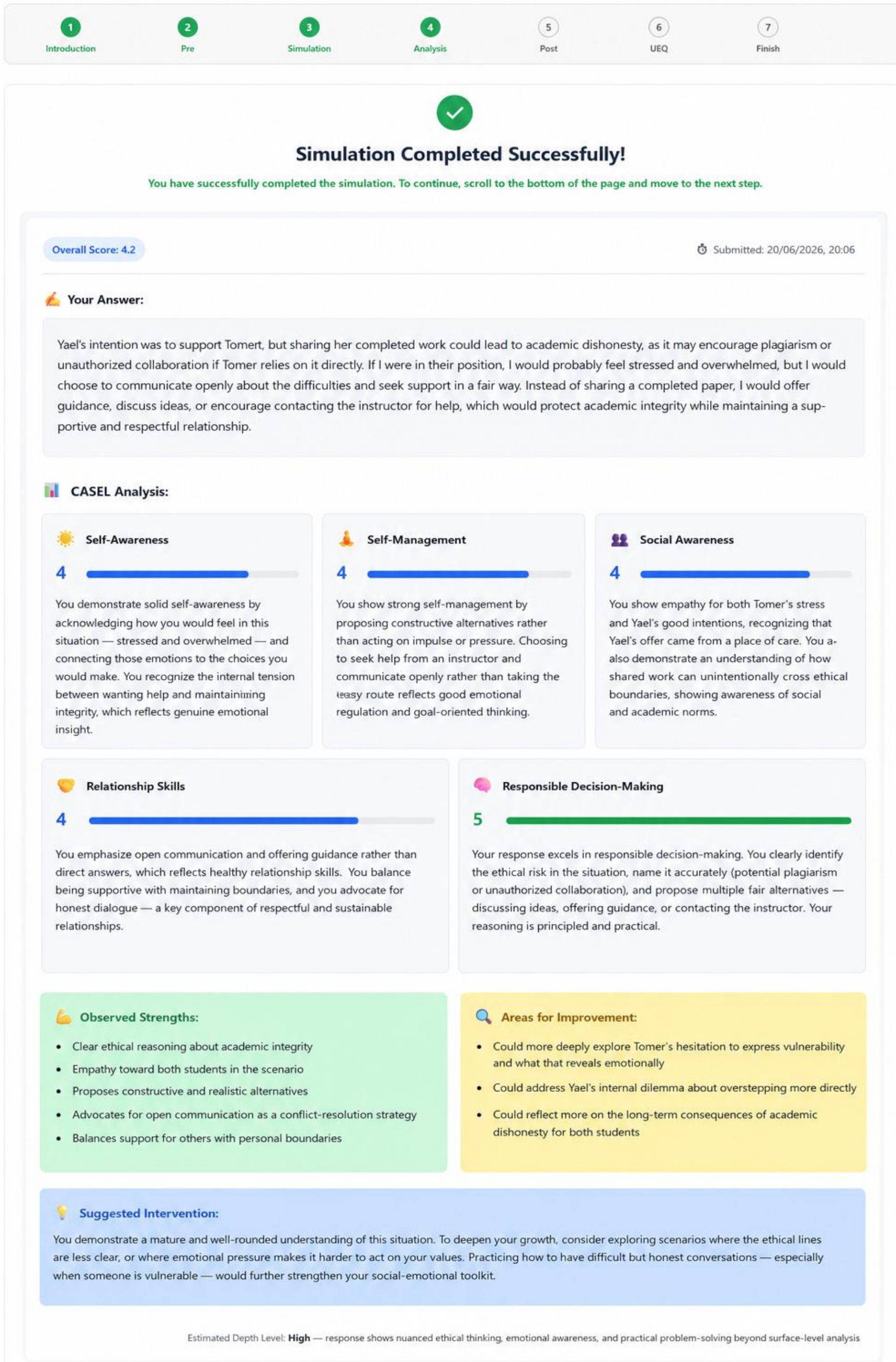

Figure 10. Analysis page

Participants assigned to the control group receive the same analysis page described above. After reviewing the feedback, participants may proceed by scrolling to the bottom of the page. At the bottom of the analysis page, a **Continue to Validated Questionnaire** button is displayed. Selecting this button transfers participants directly to the post intervention SEL questionnaire. **Figure 11** illustrates the bottom section of the analysis page presented to participants in the control group.

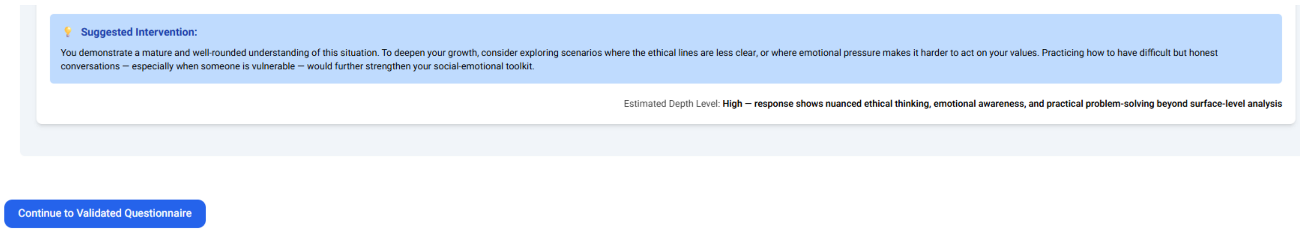

Figure 11. Bottom section of the analysis page for the control group

Participants assigned to the experimental group follow a different path after reviewing the analysis page. Instead of the **Continue to Validated Questionnaire** button presented to the control group (Figure 11), they are guided to an additional AI supported Socratic conversation. This conversation is designed to encourage reflection on the situation and is expected to last approximately eight minutes. Participants are presented with one question at a time and are invited to provide their responses in the text field before proceeding to the next question. **Figure 12** illustrates the beginning of the Socratic conversation presented to participants in the experimental group.

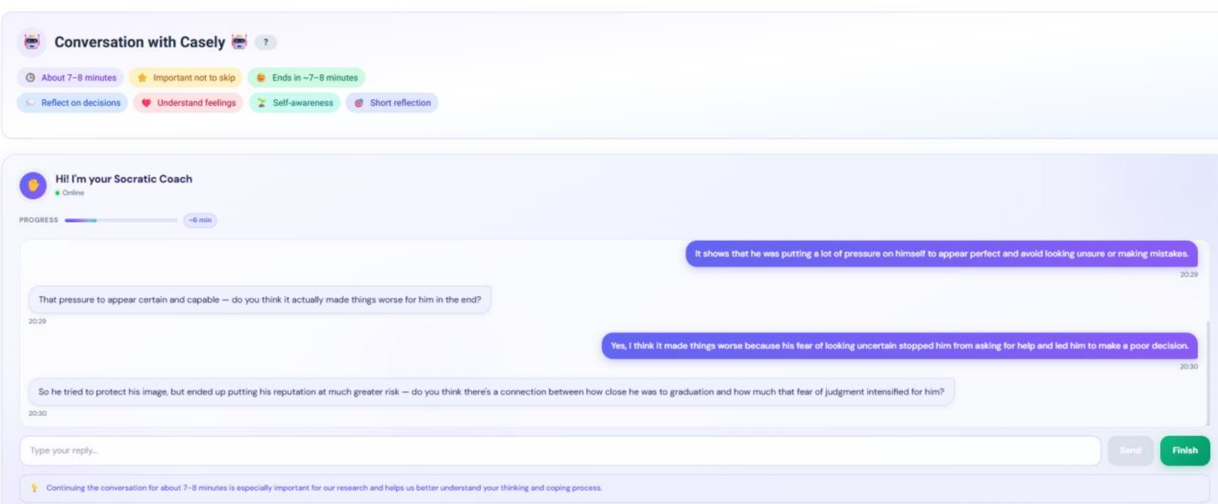

Figure 12. AI supported Socratic conversation

The following instructions apply only to participants who are presented with the AI supported Socratic conversation. During the conversation, participants are presented with one question at a time. After entering a response and selecting the **Send** button, the next question is displayed automatically. A progress indicator and an estimated remaining time are provided to help participants follow their progress throughout the conversation. Once all questions have been answered, participants may complete the discussion by selecting the **Finish** button located at the bottom right corner of the page (see Figure 12). Selecting this button transfers participants to the brief summary page, illustrated in **Figure 13**. Participants who are not presented with the AI supported Socratic conversation are transferred directly to the **post intervention SEL questionnaire**.

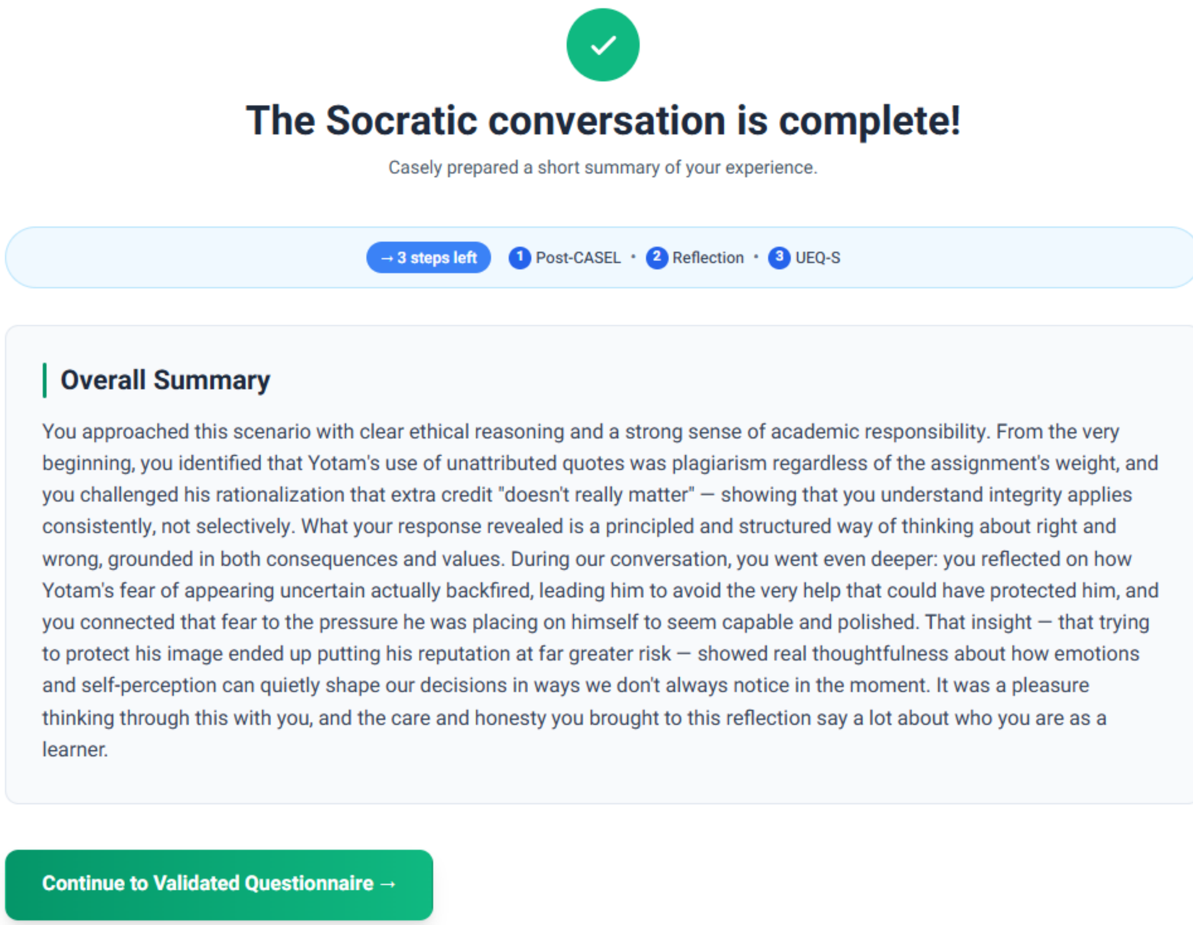

Figure 13. Brief Summary Page

This page (Figure 13) provides a concise overview of the main ideas and reflections that emerged during the discussion and displays the number of remaining stages in the study. After reviewing the summary, participants may continue by selecting the **Continue to Validated Questionnaire** button located at the bottom of the page. Selecting this button transfers participants to the post intervention SEL questionnaire.

The post intervention SEL questionnaire is accessed either by selecting the **Continue to Validated Questionnaire** button displayed at the bottom of the brief summary page for participants who completed the AI supported Socratic conversation **(see Figures 13)**, or by selecting the **Continue to Validated Questionnaire** button displayed at the bottom of the analysis page for participants who were not presented with the AI supported Socratic conversation **(see Figures 11)**. The questionnaire follows the same structure and interaction flow described previously for the pre intervention SEL questionnaire **(see Figures 5–8)**. Therefore, the questionnaire screens are not repeated in this guide. After completing the post intervention questionnaire, participants proceed to the user experience questionnaire.

After completing the post intervention questionnaire, **participants who completed the AI supported Socratic conversation** may proceed by selecting the **Finish** button located at the bottom of the questionnaire results page **(see Figure 8)**. Selecting this button transfers participants to the **Final Reflection** page, where they are invited to answer two questions about their experience with Casely. **Figure 14** illustrates the Final Reflection page. **Participants who were not presented with the AI supported Socratic conversation** are transferred directly to the user experience questionnaire after selecting the **Finish** button and may refer to **Figure 15**. After answering both reflection questions and selecting the **Finish & Continue** button, participants who completed the AI supported Socratic conversation are also transferred to the user experience questionnaire illustrated in **Figure 15**.

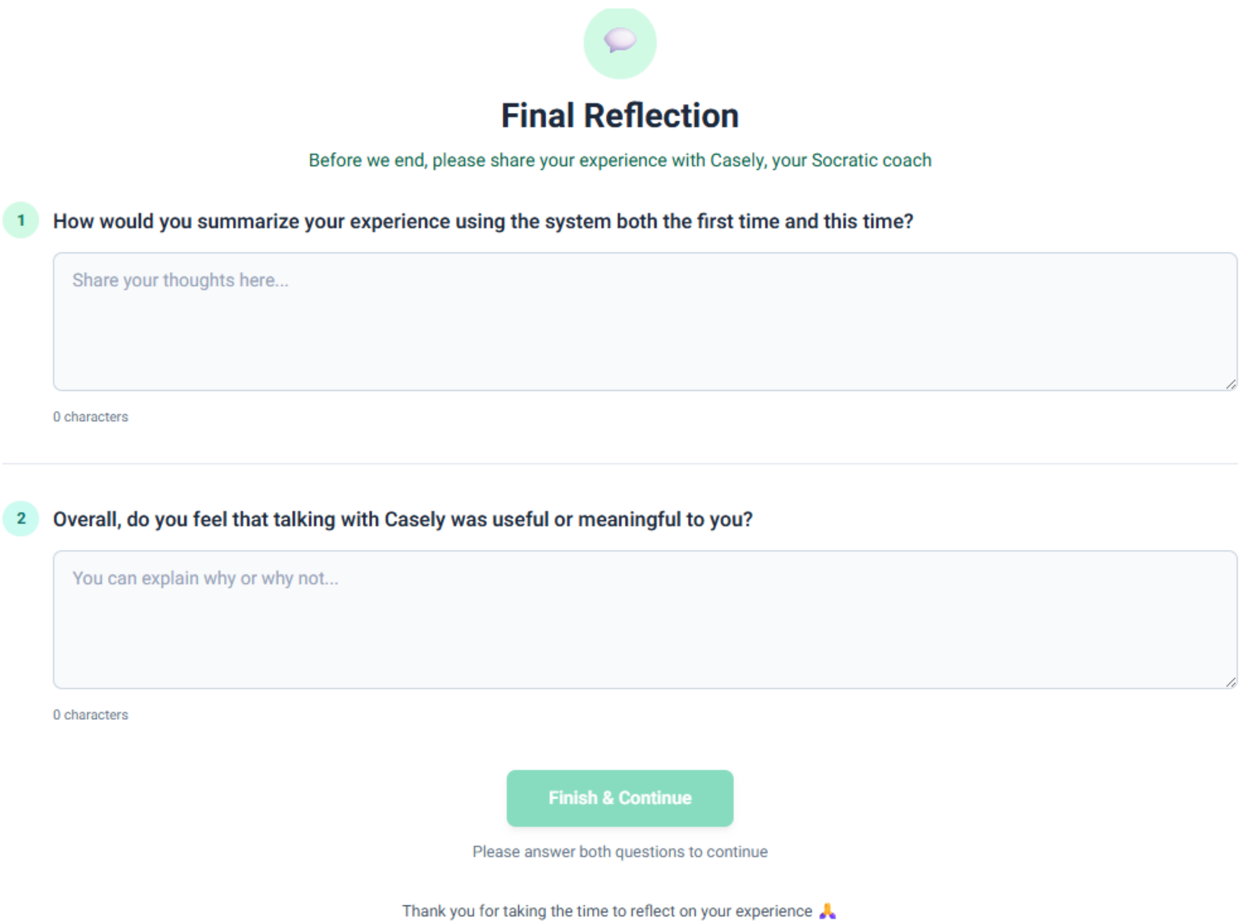

Figure 14. Final Reflection Page

Participants who are presented with the Final Reflection page are invited to answer the two reflection questions displayed in Figure 14. After entering their responses, participants may continue by selecting the **Finish & Continue** button located at the bottom of the page **(see Figure**

14). Selecting this button transfers participants to the User Experience Questionnaire (UEQ-S), illustrated in **Figure 15**.

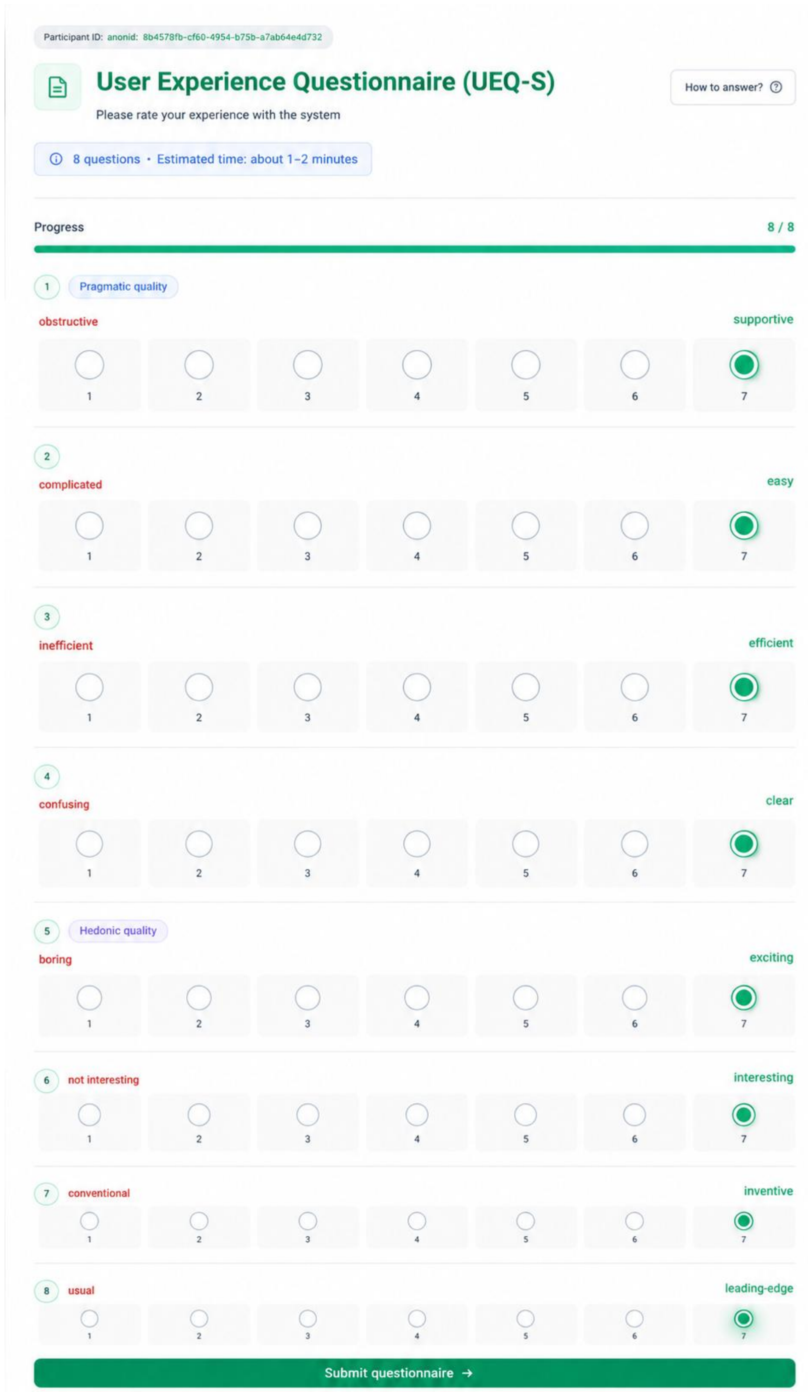

Figure 15. User Experience Questionnaire (UEQ-S) page

The User Experience Questionnaire (UEQ-S) contains eight questions. For each question, participants select one option on a seven point scale positioned between two opposite descriptions. A progress indicator displayed at the top of the page allows participants to monitor their progress throughout the questionnaire. After answering all eight questions, participants may proceed by selecting the **Submit Questionnaire** button located at the bottom of the page **(see Figure 15)**. Selecting this button transfers participants to the final page of the study, illustrated in **Figure 16**.

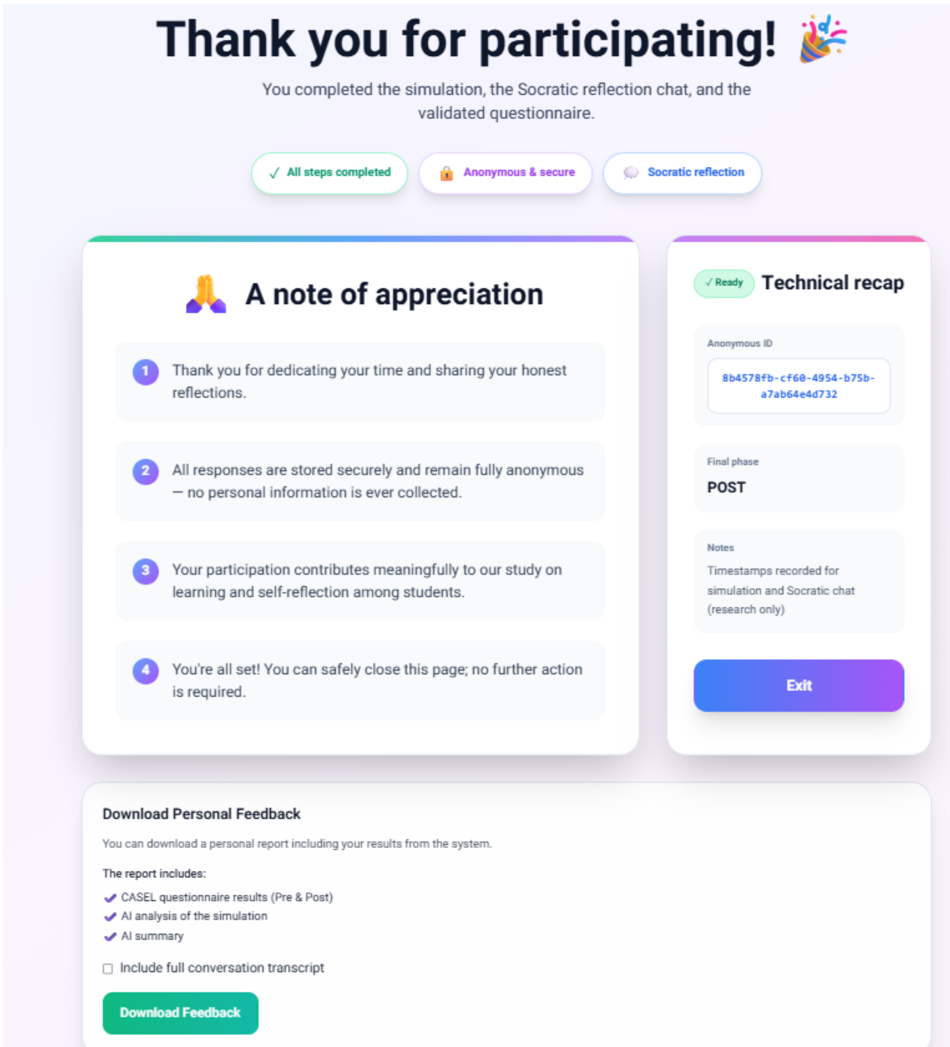

Figure 16. Study Completion Page

The study completion page confirms that all stages of the study have been completed successfully. Participants are presented with a summary message and information regarding the anonymous handling of their responses. Participants who wish to download a personal report containing their questionnaire results, simulation responses, AI analysis, and summary may do so by selecting the **Download Feedback** button located at the bottom of the page. Selecting this button downloads a PDF report to the participant's computer. Participants who wish to leave the system may select the **Exit** button located on the right side of the page. Selecting this button displays a confirmation message containing the session start time, end time, and duration.

Participants may then close the session by selecting the **Close and Continue** button. **Figure 17** illustrates the session completion confirmation message.

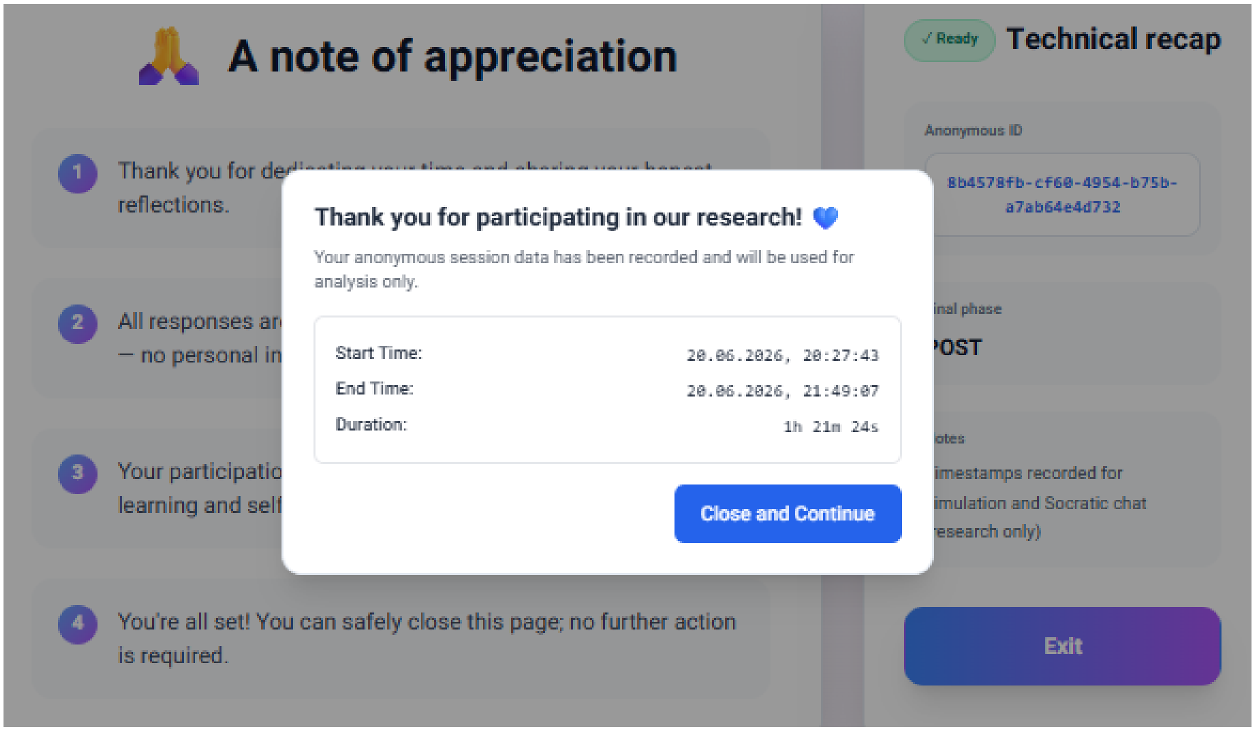

Figure 17. Session Completion Confirmation Message

The session completion confirmation message shown in **Figure 17** displays information about the completed session, including the session start time, end time, and total duration. After reviewing this information, participants may complete the process by selecting the **Close and Continue** button located at the bottom of the message window. Selecting this button closes the current session and redirects participants to the system home page, illustrated in **Figure 1**. At this point, the study workflow is complete and no further actions are required.

Completion of the Research Workflow

The sequence described in this appendix completes the entire research participation process, from entering the study to receiving the completion confirmation message and returning to the EduMap home page (see Figure 1). No further actions are required, and participants may safely close the system.

Appendix D: Maintenance Guide

1. Purpose

The purpose of this Maintenance Guide is to support the continued use and long term maintenance of the EduMap platform after the completion of the capstone project. The guide is intended for future developers and technical personnel who may need to maintain, update, or extend the system.

This document describes the operating environment of the system, including the hardware and software infrastructure required for stable operation. In addition, it presents the core technologies used by the client and server sides, the installation and configuration of the custom software developed as part of the project, and the deployment environment used in production. The guide also provides the information required to support future maintenance and system updates.

2. System Resources

The source code of the EduMap platform is maintained in the project's GitHub repository, which contains both the frontend and backend code required for future development and maintenance. The production version of the platform is accessible through the official EduMap website. These resources provide future developers with access to the application, source code, and deployment structure required for continued maintenance.

GitHub Repository:
https://github.com/NahlaAboromi/final_project

System Website:
https://edumapcasely.online/

3. System Operating Environment

3.1 Deployment and Infrastructure

The EduMap platform operates in a cloud based environment. The production system is deployed on a dedicated Contabo cloud server using an x64 based environment with 8 GB RAM. The operating system used in production is Microsoft Windows Server 2025. PM2 is used to manage the backend process and support continuous operation, while Caddy is used as a reverse proxy for HTTPS and SSL certificate management.

3.2 Client Side Environment

The client side of the system is implemented using React 19.1.0 and React DOM 19.1.0. The application uses Vite 6.3.3 as the build tool and Tailwind CSS 3.4.17 for responsive user interface styling. React Router DOM 7.5.3 is used for navigation between pages, and Recharts 2.15.3 is used for displaying charts and visual progress data.

3.3 Server Side and Data Environment

The server side is implemented using Node.js 22.14.0 with npm 11.3.0 and Express.js 5.1.0. MongoDB Atlas is used as the cloud database, and Mongoose 8.14.3 is used to define schemas and communicate with the database. The AI based functionality relies mainly on Anthropic Claude, using Claude Sonnet as the default model and Claude Haiku as a fallback model. In addition, Google Gemini models are used for specific AI based tasks such as SEL grouping and JSON repair.

4. Custom Software Installation

This section provides instructions for setting up the local development environment of the EduMap platform. These instructions are intended for development and testing purposes only. The production environment on the Contabo server is configured separately.

4. Custom Software Installation

This section describes the steps required to set up the EduMap platform in a local development environment. The source code repository contains both the frontend and backend modules. After cloning the repository from the GitHub address provided in **Section 2**, future maintainers should install the required dependencies separately for each module and then start the corresponding services. The instructions presented in this section apply only to the local host development environment used for development and testing. The deployment and configuration of the production environment on the Contabo server are described in a later section.

4.1 Local Setup and Execution

Table 1 summarizes the commands required for installing dependencies and running the system locally.

Table 1. Local Setup and Execution Commands

Component	Directory	Command
Frontend installation	hw2-frontend	npm install
Backend installation	api	npm install
Start frontend server	hw2-frontend	npm run dev
Start backend server	api	node app.js

5. Essential Configurations (Environment Variables)

The EduMap platform relies on environment variables to manage database connectivity, AI services, and email functionality. During local development, these variables are stored in a **`.env`** file located in the backend directory. Future maintainers should ensure that the required variables are correctly configured before running the application. The deployment environment on the Contabo server is configured separately and will be described in a later section.

Table 2 summarizes the main environment variables used by the system.

Table 2. Environment Variables Used in the Local Development Environment

Variable Name	Purpose
MONGO_URI	Connection string for MongoDB Atlas.
ANTHROPIC_API_KEY	Used by Claude models for analysis and Socratic dialogues.
GEMINI_API_KEY	Used for SEL grouping and JSON repair.
MAIL_USER	Gmail account used by Nodemailer.
MAIL_PASSWORD	Gmail application password used by Nodemailer.

The current **.env** configuration used for local development is summarized in **Table 3**.

Table 3. Current .env Configuration for Local Development

Variable Name	Purpose
MONGO_URI	mongodb://nahlaaboromy_db_user:B0JdRAI9WlID1sFD@ac-fcwewn1-shard-00-00.snlp5m6.mongodb.net:27017,ac-fcwewn1-shard-00-01.snlp5m6.mongodb.net:27017,ac-fcwewn1-shard-00-02.snlp5m6.mongodb.net:27017/modular_skills?ssl=true&replicaSet=atlas-846hkp-shard-0&authSource=admin&retryWrites=true&w=majority&appName=Cluster0
ANTHROPIC_API_KEY	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
GEMINI_API_KEY	AQ.Ab8RN6JECyTUzt2SsJXQ1xLoq_0DEhui4LTvO5Bt51jvmIndbg
MAIL_USER	n0502898789@gmail.com
MAIL_PASSWORD	btcvlusngxfnaiov

6. Database Structure and Maintenance

Using the MongoDB Atlas connection string (MONGO_URI) specified in **Table 3**, persistent data is stored in MongoDB Atlas. The research process relies on several primary collections.

anonymous_students: Stores anonymous participant identifiers and demographic information collected during the research process.

trials: Stores participant group assignments, simulation responses, AI analysis results, chat history, interaction statistics, and final reflections.

sel_assessments: Stores pre and post CASEL questionnaire responses together with completion timestamps.

ueq_assessments: Stores UEQ-S responses and the calculated pragmatic, hedonic, and overall scores.

scenarios: Stores simulation scenarios and their associated questions used throughout the study.

Future maintainers should preserve the relationships between these collections through the anonymous identifier (anonId), which serves as the primary link between participant information, simulation data, SEL assessments, and UEQ-S evaluations. Database administration and backup procedures are managed through MongoDB Atlas.

7. Production Deployment Environment

The production environment of the EduMap platform is hosted on a dedicated Contabo cloud server running Microsoft Windows Server 2025. Unlike the local host development environment described in Sections 4 and 5, the production environment is configured directly on the server.

The backend service is managed using PM2, which ensures continuous execution and automatic recovery after server restarts. Caddy is used as a reverse proxy and automatically manages HTTPS certificates to provide secure communication. External traffic is received through ports 80 and 443, while the backend API runs internally on port 5000.

Future maintainers should ensure that PM2 and Caddy services are active after system updates or server restarts. The production environment uses its own configuration and environment variables, which are maintained separately from the local development environment.

7.1 Production Deployment Steps

The first step in deploying the EduMap platform is to create a cloud server through the Contabo website: https://contabo.com/en-us/vds/

As shown in **Figure 18**, the Contabo website provides several cloud server configurations. The maintainer should select and purchase a Windows Server 2025 based cloud server.

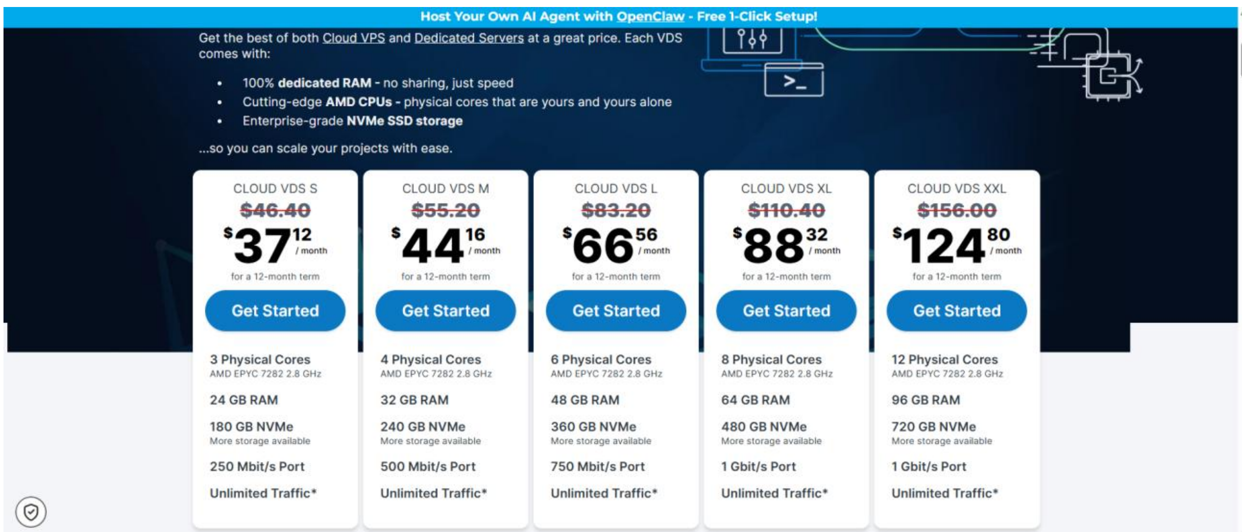

Figure18. Contabo cloud server selection page.

The deployment process is straightforward. After purchasing the server through the Contabo website, one of the most important steps is to provide the **email address and password** that will be associated with the server account. Once the order is completed, Contabo sends an email containing the server details, including the public IP address, server type, user name, and password required for remote access. **Figure 19** presents an example of the information received after the server is created. These details will be used in the next step to connect to the server remotely.

Your VPS

IP address	server type	Location	user name	password
185.135.137.193	Cloud VPS 10 NVMe (no setup)	Hub Europe	root	as chosen by you during order process

Figure 19. Example of server information provided by Contabo after server creation.

After receiving the server information shown in **Figure 19**, the maintainer can access the production server remotely using the Remote Desktop Connection utility provided by Microsoft Windows. The application can be easily located through the Windows search menu, as shown in **Figure 20.**

Figure 20. Accessing the Remote Desktop Connection utility from the Windows search menu.

After locating and opening the **Remote Desktop Connection** utility shown in **Figure 20**, the initial connection window shown in **Figure 21** will appear. The maintainer should enter the **public IP address** received from Contabo, as presented in **Figure 19**, and then click Connect.

Figure 21. Initial Remote Desktop Connection window.

After entering the public IP address and clicking Connect, the Windows Security window shown in **Figure 22** will appear. The maintainer should enter the username and password associated with the server account that were defined during the server purchase process. Once the credentials are entered successfully, the remote connection to the production server will be established.

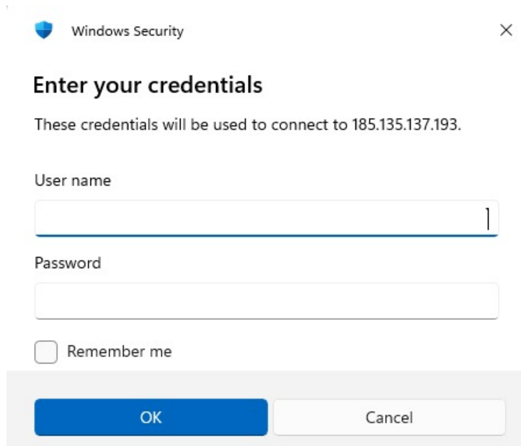

Figure 22. Windows Security authentication window for accessing the Contabo server.

After entering the username and password in the Windows Security window shown in **Figure 22**, the Remote Desktop Connection window will reappear with the server information filled in automatically. As shown in **Figure 23,** the maintainer should verify the public IP address and user name and then click Connect to establish the remote connection to the Windows Server 2025 environment.

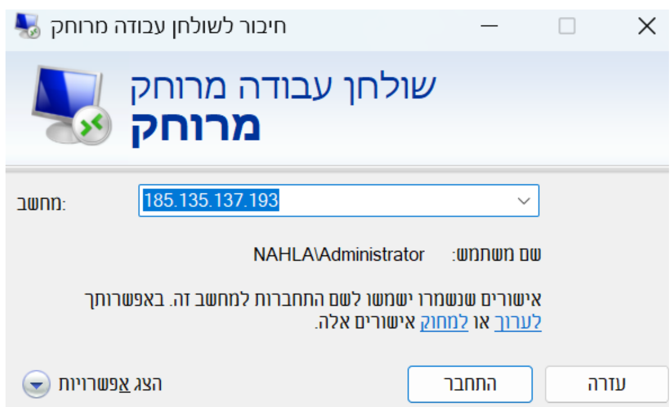

Figure 23. Final Remote Desktop Connection window before accessing the Windows Server 2025 environment.

After clicking Connect in the Remote Desktop Connection window shown in **Figure 23**, the remote connection to the server will be established successfully. The Windows Server 2025 desktop will then appear, as shown in **Figure 24**. From this environment, the maintainer can access the server files and continue with the deployment and configuration of the EduMap platform.

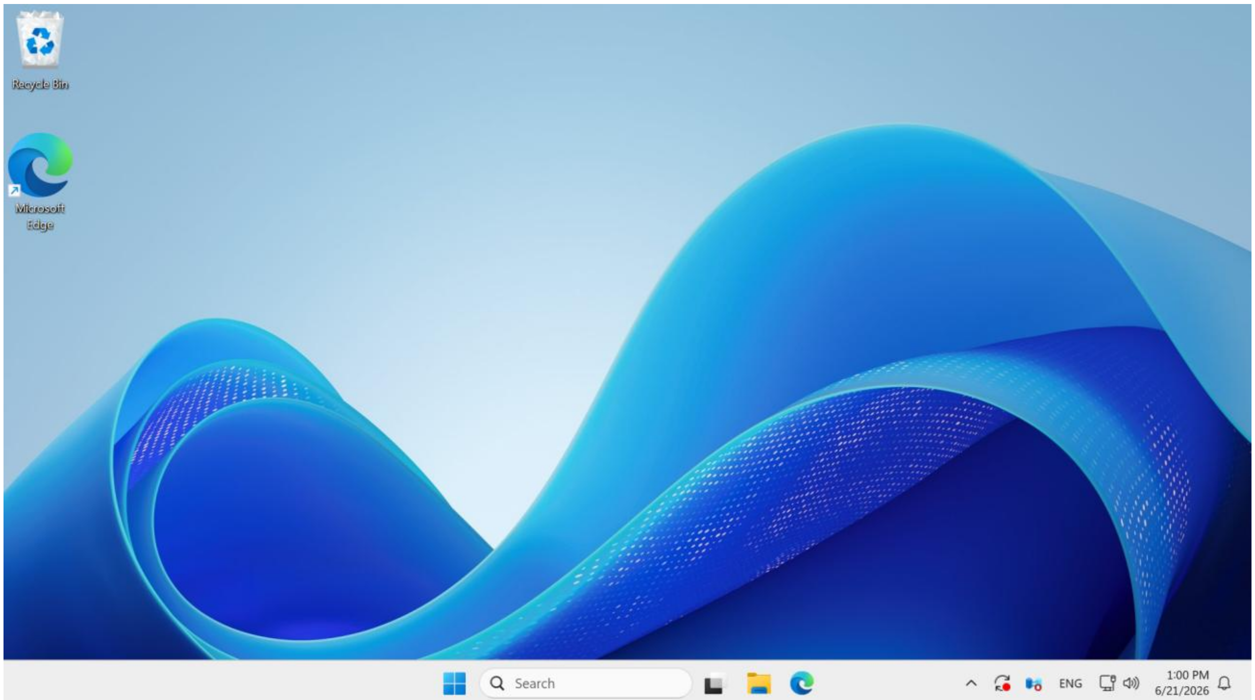

Figure 24. Windows Server 2025 desktop after successful remote connection.

After establishing the remote connection shown in **Figure 24**, the maintainer should prepare the Windows Server 2025 environment for deploying the EduMap platform. The required software components and system configurations are identical to those described previously in Sections 3, 4, and 5. This includes installing the required tools, cloning the GitHub repository, configuring the environment variables, and installing the frontend and backend dependencies. Once these steps are completed, the system is ready for execution and production deployment.